HSPT Exam Simulator

Full-Length Practice Exam 15

Complete printable test with 298 multiple-choice questions, answer key, and detailed explanations.

Section	Questions	Time
Verbal Skills	60	20 minutes
Quantitative Skills	52	35 minutes
Reading Comprehension	62	35 minutes
Mathematics	64	45 minutes
Language	60	30 minutes

General Directions

Work through each section within the recommended time. Choose the best answer for every question. Mark one answer (A-D) for each item. Use the answer key and explanations only after completing the test.

Recommended total testing time: 165 minutes. Calculators should be used only when permitted by the rules followed for your HSPT preparation.

Section 1: Verbal Skills

60 questions | Recommended time: 20 minutes

1. After the long climb, the hikers had to TRUDGE through the last stretch. TRUDGE most nearly means:

A) run with quick, light steps

B) slide smoothly without effort

C) walk with heavy, weary steps

D) turn suddenly toward another path

2. Faced with two equally troublesome choices, Ellis was in a QUANDARY. QUANDARY most nearly means:

A) a state of difficulty deciding what to do

B) a period of complete agreement

C) a public celebration of success

D) a formal promise to keep silent

3. The critic dismissed the argument as SOPHISTRY. SOPHISTRY most nearly means:

A) evidence that conclusively settles a dispute

B) a brief statement of widely accepted facts

C) an openly admitted lack of an argument

D) reasoning that sounds convincing but is misleading or unsound

4. UPSTREAM means toward a river's source. Which direction is its opposite?

A) toward the river's source

B) across the river from bank to bank

C) along a path away from either bank

D) toward the river's mouth

5. Which description is most nearly opposite to DOG-EARED, describing pages after long use?

A) having folded-down corners

B) being well-thumbed and creased

C) having flat, uncreased corners

D) being worn and curled at the corners

6. Which word is most nearly opposite to SANGUINE in "sanguine about the outcome"?

A) hopeful

B) confident

C) optimistic

D) pessimistic

7. A crushable packing insert is designed to deform before an impact damages the packed object. A deliberately weak breakaway link in machinery is similarly designed to:

A) remain rigid while transmitting excessive force to the connected parts

B) break before excessive force damages more valuable parts

C) break only after the parts it protects have already failed

D) make the more valuable connected parts yield before it does

8. A plural noun ending marks NUMBER. A past-tense verb ending similarly marks:

A) ownership

B) time

C) direction

D) physical size

9. A letter transposition swaps letters in writing. A spoonerism such as "cold toast" becoming "told coast" similarly swaps:

A) the meanings of the two words

B) initial sounds between words

C) the grammatical tenses of verbs

D) the final syllables of words

10. Which time marker depends on a natural cycle rather than a human arrangement?

A) an appointment time

B) a train's scheduled departure

C) sunrise

D) an announced submission deadline

11. Which set contains ONLY expressions whose reference normally depends directly on who is speaking and where or when the words are spoken?

A) I; here; today

B) she; triangle; today

C) we; London; now

D) here; oxygen; you

12. Which pair of descriptions cannot both be true but can both be false for the same allowed case?

A) positive; negative, for a real number

B) zero; nonzero, for a real number

C) even; odd, for an integer

D) equal in length; unequal in length, for two segments

13. A speaker claims, "If we permit one late return, soon every deadline will disappear and no rule will be followed." No intermediate evidence is offered. What is the main weakness?

A) The speaker has shown that every exception immediately improves a rule.

B) The speaker treats an unsupported chain of increasingly extreme consequences as inevitable.

C) The claim is invalid solely because it mentions the future.

D) The claim proves that the first late return has already occurred.

14. A traveler starts outside an enclosure, enters it at least once, and finishes outside. Every crossing uses either the north gate or the south gate and is recorded. Which statement must be true?

A) The total number of entries equals the total number of exits.

B) Each gate must be used equally often for entry and exit.

C) Every entry must use a different gate from the following exit.

D) The north gate must be used more often than the south gate.

15. A pass permits its holder to attend a morning session, an evening session, or both. Nothing is said about what a particular holder chose. Which conclusion is justified?

A) The holder necessarily attended both sessions.

B) The holder necessarily attended exactly one session.

C) Permission alone does not establish which sessions the holder actually attended.

D) The holder was forbidden to attend the evening session.

16. After several sleepless nights at sea, the sailor looked HAGGARD. HAGGARD most nearly means:

A) cheerful and refreshed

B) angry and impatient

C) elegant and carefully dressed

D) worn and exhausted

17. The metal is MALLEABLE enough to be shaped with a hammer. MALLEABLE most nearly means:

A) likely to shatter when struck

B) unable to conduct heat

C) difficult to separate from a magnet

D) capable of being pressed into a new shape

18. Miri tried to CAJOLE her reluctant cousin into joining the dance. CAJOLE most nearly means:

A) persuade with coaxing or flattery

B) force by threatening punishment

C) prevent by placing an obstacle

D) confuse with unrelated information

19. Which word is most nearly opposite to BRUSQUE, meaning abrupt or blunt in manner?

A) curt

B) courteous

C) gruff

D) impatient

20. The committee gave the proposal only TEPID support. Which word most nearly expresses the opposite of TEPID here?

A) halfhearted

B) lukewarm

C) uninterested

D) enthusiastic

21. A FICKLE admirer quickly switches loyalties. Which word most nearly expresses the opposite of FICKLE?

A) steadfast

B) changeable

C) unpredictable

D) inconsistent

22. A stencil guides painting by keeping paint off selected parts of a surface. A mold similarly guides casting by:

A) changing the color of the material

B) measuring the weight of the finished object

C) constraining material within a chosen shape

D) polishing the object after it hardens

23. A written musical rest specifies a period of silence between sounds. Which feature of ordinary printed prose has the most similar role?

A) A capital letter beginning a name

B) A space separating neighboring words

C) A page number locating a passage

D) An italic letter emphasizing a word

24. An abbreviation leaves out letters while preserving a recognizable reference to a word. Which action has the most similar relationship to its original?

A) Replacing a damaged object with an unrelated object

B) Arranging the same objects from largest to smallest

C) Copying every mark in an original drawing

D) Drawing a recognizable outline while leaving out an object's fine details

25. Which word belongs with HINGE, CLASP, and LATCH as a piece of fastening or connecting hardware?

A) bevel

B) bolt

C) groove

D) scratch

26. Which word does NOT belong with the others as a verb that can grant permission?

A) authorize

B) applaud

C) permit

D) license

27. A THRESHOLD marks an entrance boundary, a BORDER separates regions, and a SEAM joins edges. Which word also names a place where distinct parts meet?

A) interior

B) center

C) junction

D) core

28. A package receives a blue tab only after its address is checked. Removing the tab does not undo the check. A package has no blue tab now. Which conclusion is justified?

A) Its address has never been checked.

B) Its address must already have been checked.

C) Its address may have been checked, but the missing tab does not establish that.

D) Its address cannot be checked in the future.

29. Four film clips, Drift, Harbor, Orchard, and Summit, appear once each. Drift is immediately before Harbor. Orchard is after Harbor but before Summit. Which placement is forced?

A) Summit is immediately before Drift.

B) Harbor appears after Orchard.

C) Orchard is immediately after Harbor.

D) Drift appears immediately before Summit.

30. Two witnesses are either both truthful or both untruthful. One says, "The latch is brass." The other says, "The latch is brass and the door is blue." The first statement is false. What must follow?

A) The latch is not brass, but the door's color remains undetermined.

B) The door must be blue.

C) The door cannot be blue.

D) The latch must be brass because both witnesses agree.

31. The room's dim light made the actor's face appear PALLID. PALLID most nearly means:

A) sunburned

B) pale

C) cheerful

D) furious

32. Neither side would change its proposal, and the talks reached an IMPASSE. IMPASSE most nearly means:

A) an agreement accepted eagerly

B) a meeting held without notice

C) a point at which progress is blocked

D) a proposal requiring little effort

33. Without maintenance or new customers, the once-busy workshop began to LANGUISH. LANGUISH most nearly means:

A) expand with sudden success

B) move to a larger building

C) operate with unusual precision

D) grow weak or fail to prosper

34. Which word is most nearly opposite to COARSE when describing the texture of fabric?

A) rough

B) scratchy

C) harsh

D) fine

35. The crew offered a TENTATIVE date that might change. Which word most nearly expresses the opposite of TENTATIVE here?

A) provisional

B) uncertain

C) definite

D) temporary

36. The report was REPLETE WITH examples. Which phrase most nearly expresses the opposite of REPLETE WITH?

A) devoid of

B) full of

C) rich in

D) packed with

37. A map legend explains the symbols on a map. Which has the most similar relationship?

A) A dictionary cover protects the pages inside it.

B) A pronunciation key explains the sound symbols in a dictionary.

C) A map scale compares map distance with actual distance.

D) A compass points toward a particular direction.

38. A rehearsal exposes difficulties before a performance is presented. A prototype serves the most similar purpose when it:

A) reveals design problems before a product is manufactured widely

B) advertises a product after all manufacturing has ended

C) records the names of everyone who bought a product

D) stores finished products while they await shipping

39. A correction changes the faulty part of a claim, while a retraction withdraws the claim. Which pair makes the most similar distinction?

A) Repairing a device carefully; repairing it quickly

B) Repairing a damaged device; discarding the device entirely

C) Buying a device; lending it to someone

D) Moving a device upstairs; moving it downstairs

40. Which item is an assigned identifier rather than a measurement of a physical property?

A) A parcel's weight

B) A parcel's height

C) A parcel's temperature

D) A parcel's tracking code

41. Which group contains ONLY activities that ordinarily involve an exchange of information between participants?

A) dialogue; discussion; negotiation

B) dialogue; broadcast; negotiation

C) announcement; discussion; monologue

D) broadcast; announcement; monologue

42. A spare tire, emergency fuel, and a backup fund share which function?

A) They are residues left after a process finishes.

B) They are wastes that can no longer be used.

C) They are reserves kept for a possible future need.

D) They are rewards distributed after a task succeeds.

43. A design is approved if the curator approves it alone OR if both the designer and carpenter approve it. Those are the only approval routes. The designer approves, but the other two do not. What follows?

A) The designer's approval is sufficient by itself.

B) The carpenter's refusal cancels the curator's approval.

C) The design is approved because someone supports it.

D) The design has not met either approval route.

44. Three statements describe the same moment: "Gate A is open"; "Gate B is open"; "Both gates are open." Exactly one statement is true. Which conclusion must be true?

A) Exactly one of the two gates is open.

B) Both gates are open.

C) Both gates are closed.

D) Gate A is open and Gate B is closed.

45. Whenever a ventilation fan starts, a display lamp goes out. An observer concludes that the fan causes the lamp to go out. Which finding most weakens that conclusion?

A) The fan makes a noticeable sound when starting.

B) The lamp has a green cover.

C) A timer independently starts the fan and switches off the lamp at the same moment.

D) The fan's housing is larger than the lamp's housing.

46. At first light, the shorebirds began to FORAGE along the exposed mud. FORAGE most nearly means:

A) prepare to migrate

B) hide from the wind

C) make a warning sound

D) search for food

47. Even when others withdrew, Imani remained a STAUNCH supporter of the proposal. STAUNCH most nearly means:

A) easily confused

B) quietly resentful

C) firmly loyal

D) reluctantly involved

48. The young architect proposed an AUDACIOUS design that surprised the cautious panel. AUDACIOUS most nearly means:

A) boldly daring

B) strictly conventional

C) barely visible

D) carelessly unfinished

49. Which word is most nearly opposite to WARY, meaning alert to possible danger?

A) incautious

B) watchful

C) guarded

D) vigilant

50. The workers described the assignment as ONEROUS because it demanded so much effort. Which word most nearly expresses the opposite of ONEROUS?

A) burdensome

B) undemanding

C) exhausting

D) arduous

51. The inspector dismissed the scratch as a TRIVIAL defect. Which word most nearly expresses the opposite of TRIVIAL here?

A) minor

B) negligible

C) insignificant

D) consequential

52. A knitting pattern specifies how stitches are arranged. Which has the most similar relationship?

A) A costume protects a dancer from cold weather.

B) Choreography specifies how dance movements are arranged.

C) A curtain prevents an audience from seeing a stage.

D) A ticket records permission to attend a performance.

53. A dimmer adjusts a working lamp's brightness. Which action relates to a running engine in the most similar way?

A) Draining its fuel until it stops

B) Adjusting its throttle to change its output

C) Repainting its outer cover

D) Removing it from the vehicle for replacement

54. A printing line cannot produce finished books faster than its slow binding station, even when the presses run faster. Which situation follows the same principle?

A) An unused parking area becomes emptier after a road closes.

B) A road sign changes color without changing the road.

C) A narrow passage limits traffic arriving from several wide roads.

D) A passenger chooses a different seat on an uncrowded bus.

55. A dent, a footprint, and a stamped impression belong together because they are:

A) measurements of brightness

B) names for weather conditions

C) sounds made by vibrating strings

D) marks left by physical contact

56. Which set contains ONLY descriptions of whether a resource is available for use?

A) vacant; occupied; reserved

B) vacant; polished; reserved

C) heavy; occupied; bright

D) reserved; wide; polished

57. Which document aims to preserve the original wording rather than condense its main ideas?

A) An abstract

B) A synopsis

C) A verbatim transcript

D) An overview

58. Before a boat launches, both its crew check and radio check must be complete. A fuel check must precede the crew check. No other order is required. Which statement must be true?

A) The radio check must precede the fuel check.

B) The fuel check can wait until after launch.

C) The fuel check occurs before launch, but its order relative to the radio check is not fixed.

D) The crew check must occur before the radio check.

59. Every northbound vehicle is electric or has a permit. No electric vehicle is a truck. A log lists a northbound truck without a permit. If both rules are correct, what must follow?

A) The log proves that all trucks are electric.

B) At least one description in that log entry is incorrect.

C) The truck qualifies because northbound travel needs no permit.

D) The rules allow a non-electric truck to travel north without a permit.

60. Every tile is either glazed or unglazed, never both. Rule 1 says all glazed tiles are smooth. Rule 2 says all unglazed tiles are rough. Which statement about disproving these rules is correct?

A) One rough glazed tile would disprove both rules.

B) One smooth unglazed tile would disprove both rules.

C) One rough unglazed tile would disprove Rule 2.

D) Disproving both rules would require at least two tiles, one from each group.

Section 2: Quantitative Skills

52 questions | Recommended time: 35 minutes

61. Term n counts how many times the positive integer n can be halved while remaining an integer before an odd integer is reached.
For n = 1 through 8, the terms are 0, 1, 0, 2, 0, 1, 0, 3. What is term 16?

A) 4

B) 3

C) 5

D) 16

62. A queue starts empty. In successive minutes, 3, 1, 4, 1, and 5 items arrive. After each arrival batch, up to 2 waiting items are processed; unused capacity is not saved. The end-of-minute queue lengths begin 1, 0, 2, 1. What is the fifth length?

A) 3

B) 6

C) 0

D) 4

63. Each term is a three-digit string of zeros and ones. To obtain the next term, discard the first digit and shift the other two left. Append 0 if the ORIGINAL first and last digits match, or 1 if they differ.
101, 010, 100, 001, 011, 111, ___

A) 101

B) 110

C) 011

D) 111

64. Term n counts circular patterns of n beads, each black or white. Patterns that match after rotating the circle count as the same; the circle has no marked starting bead. The first three terms are 2, 3, and 4. What is the fourth term?

A) 5

B) 8

C) 6

D) 16

65. Let min(a, b) mean the smaller value and max(a, b) the larger value. What is max(−7, −2) − min(5, 8)?

A) −7

B) −12

C) 3

D) −10

66. A code splits six binary digits into two groups of three. Within each group, the left, middle, and right digits have weights 4, 2, and 1; add the weights where the digit is 1. The two results become a two-digit code. What code comes from 101110?

A) 53

B) 65

C) 11

D) 56

67. How many pairs of positive integers a and b with a < b satisfy 1/a + 1/b = 1/6?

A) 2

B) 4

C) 5

D) 8

68. The real numbers a, b, and c have mean m, where m ≠ 0.
Quantity A: $(a - m)^2 + (b - m)^2 + (c - m)^2$
Quantity B: $a^2 + b^2 + c^2$

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

69. The numbers a and b are real.
Quantity A: $|ab|$
Quantity B: $|a| \times |b|$

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

70. A finite list of positive numbers includes repeated values. A new list keeps each distinct value exactly once.
Quantity A: The mean of the new list
Quantity B: The mean of the original list

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

71. A convex polygon has n sides, where n ≥ 4.
Quantity A: The number of triangles formed by joining one interior point to every vertex
Quantity B: The number of triangles formed by drawing all possible diagonals from one vertex

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

72. Two adjacent angles form a straight angle. Each is bisected.
Quantity A: The smaller angle between the two bisecting rays
Quantity B: 90°

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

73. A square and a circle have the same positive perimeter.
Quantity A: The area of the square
Quantity B: The area of the circle

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

74. To obtain each next entry, reverse the two digits, then subtract the smaller of the original and reversed numbers from the larger. Keep a leading zero when needed. What comes next?
83, 45, 09, 81, 63, ...

A) 18

B) 36

C) 27

D) 54

75. Each entry after the first two is the remainder when the earlier of the preceding two entries is divided by the later. What comes next?
83, 34, 15, 4, 3, ...

A) 1

B) 0

C) 2

D) 12

76. Start with 1. To get the next term, add 2 to the current term, then divide that result by the current term plus 1. What follows?
1, 3/2, 7/5, 17/12, ...

A) 29/41

B) 19/13

C) 31/24

D) 41/29

77. The block 246 repeats without stopping in 0.246246246... . What is the seventeenth digit to the right of the decimal point?

A) 2

B) 4

C) 6

D) 7

78. An integer n is greater than 70 and less than 100. It is divisible by 9, and n + 1 is divisible by 5. What is n?

A) 72

B) 81

C) 99

D) 90

79. Use ordinary rounding, with a halfway value rounded up. Which is the smallest positive number with exactly two decimal places that rounds to 4 as a whole number and to 3.5 to the nearest tenth?

A) 3.50

B) 3.45

C) 3.49

D) 3.55

80. How many triples of positive integers (a, b, c) satisfy $a < b < c$ and $a + b + c = 12$?

A) 3

B) 7

C) 6

D) 12

81. The number x satisfies $0 < x < 1$.
Quantity A: $(1 + x)/(1 - x)$
Quantity B: $1 + 2x$

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

82. A positive integer k is one more than a multiple of 4.
Quantity A: The remainder when k^2 is divided by 8
Quantity B: 3

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

83. Three nonzero real numbers x, y, and z satisfy $x + y + z = 0$.
Quantity A: xyz
Quantity B: 0

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

84. Point D is the midpoint of side BC of triangle ABC. Segment AD is drawn.
Quantity A: The area of triangle ABD
Quantity B: The area of triangle ACD

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

85. A cylinder and a cone have the same circular base radius and the same height.
Quantity A: The cylinder's volume
Quantity B: Three times the cone's volume

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

86. Two sides of a triangle measure 5 and 7 units. Their included angle is not specified.
Quantity A: The triangle's area in square units
Quantity B: 17.5

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

87. The entries count the letters in the English names of successive positive integers, beginning with ONE. What comes next?
3, 3, 5, 4, 4, 3, ...

A) 3

B) 5

C) 4

D) 7

88. To obtain the next term, add the sum of the current term's digits to that term. What follows?
19, 29, 40, 44, 52, ...

A) 59

B) 57

C) 60

D) 104

89. Each term after the first three equals the arithmetic mean of the preceding three terms. What follows?
2, 4, 6, 4, 14/3, ...

A) 14/3

B) 22/3

C) 14/9

D) 44/9

90. The entries count ordered ways to travel exactly n units using steps of 1 or 3 units, for n = 1, 2, 3, 4, 5, ... What comes next?
1, 1, 2, 3, 4, ...

A) 5

B) 7

C) 6

D) 8

91. The digits 0, 3, and 6 must each be used exactly once. What is the greatest three-digit odd integer that can be formed?

A) 603

B) 630

C) 360

D) 306

92. What is the smallest positive integer k that is divisible by 2 and makes 45k a perfect square?

A) 5

B) 20

C) 10

D) 40

93. Choose two different nonzero digits, form a two-digit number, and add the number obtained by reversing its digits. How many distinct sums are palindromes, meaning they read the same forward and backward?

A) 7

B) 9

C) 16

D) 8

94. The numbers a and b both exceed 1.
Quantity A: ab
Quantity B: a + b − 1

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

95. The six observations are a, b, c, −a, −b, and −c, where a, b, and c are positive.
Quantity A: The median of the six observations
Quantity B: 0

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

96. Events E and F each have probability 1/2. Their relationship is otherwise unspecified.
Quantity A: The probability that both E and F occur
Quantity B: 1/4

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

97. A square has positive side length s and diagonal length d.
Quantity A: d − s
Quantity B: s

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

98. A cube has edge length s.
Quantity A: The combined area of its four faces other than the top and bottom
Quantity B: The area of a square with side length 2s

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

99. A rectangle measures 4 by 1 units. A triangle has a base of 4 units and perpendicular height of 2 units, but its apex may lie anywhere at that height.
Quantity A: The triangle's perimeter
Quantity B: The rectangle's perimeter

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

100. A dial has positions 0 through 5 arranged in that order around a circle. Start at 0, move forward 1 position, then 2, then 3, and continue increasing the move by 1. What is the next recorded position?
0, 1, 3, 0, 4, ...

A) 3

B) 5

C) 9

D) 2

101. Multiply each term by 1.5, then round to the nearest whole number, rounding halfway values up. What follows?
4, 6, 9, 14, 21, ...

A) 42

B) 31

C) 31.5

D) 32

102. Begin with 1. To obtain each next entry, multiply the current entry by 10 and take the remainder after division by 7. The first entries are 1, 3, 2, 6, 4, 5. What is the twentieth entry?

A) 1

B) 3

C) 2

D) 5

103. Folders are labeled with successive positive integers, but any label containing the digit 4 is skipped. What label is placed on the tenth folder?

A) 9

B) 10

C) 11

D) 12

104. Machine A doubles a nonzero input and then squares the result. Machine B squares the same input and then doubles the result. Machine A's output is how many times Machine B's output?

A) 1

B) 4

C) It varies with the nonzero input.

D) 2

105. Let x = 0.181818... , with the block 18 repeating forever. What is the exact value of 11x?

A) 2

B) 1.98

C) 1.998

D) 2.2

106. Two positive integers have sum 13, and their product is divisible by 12. What is the greatest possible product?

A) 12

B) 36

C) 40

D) 42

107. The number x is real.
Quantity A: $(x + 1)^2 - (x - 1)^2$
Quantity B: 4x

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

108. The integers a and b satisfy 0 < a < b.
Quantity A: b/a − 1
Quantity B: 1 − a/b

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

109. The integer n is greater than 1.
Quantity A: The arithmetic mean of 1, 2, 3, ..., n
Quantity B: n

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

110. Two rectangular boxes have equal positive volumes, but their dimensions may differ.
Quantity A: The total length of the first box's twelve edges
Quantity B: The total length of the second box's twelve edges

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

111. A cube has positive edge length.
Quantity A: A diagonal connecting opposite vertices through the cube's interior
Quantity B: A diagonal lying on one square face

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

112. A regular hexagon is inscribed in a circle of radius r.
Quantity A: The hexagon's perimeter
Quantity B: 6r

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

Section 3: Reading Comprehension

62 questions | Recommended time: 35 minutes

Literary Passage: The Jay's Moving Marker

The jay pushed a nut into the soft ground at the tip of a post's long shadow. It looked from the post to the dark line and back again. Here was a useful arrangement: a black finger pointing to exactly one patch of earth. The bird tapped the tip twice, as though thanking it, and flew away.

When it returned, the sun was higher and the shadow shorter. The post stood where it had always stood. The jay followed its finger to the new tip and dug. It found a pale grub, a curled leaf, and no nut.

The bird dug more vigorously. Then it stopped to inspect the post, first with one eye and then with the other. There was nothing unusual about it. The black finger still looked straight and certain.

A little farther away, beside a band of yellow leaves, the jay noticed soil that had already been disturbed. It scratched there and uncovered the nut. The shadow's tip lay several hops away from the hole.

For a moment the bird stood between its food and the helpful-looking line. Then it carried the nut to a root scored with two deep notches and buried it beside the larger notch. It gave the post one short, indignant cry before leaving.

The root offered no directions. It simply stayed.

113. What does the jay first use to mark the nut's location?

A) The larger notch in a root

B) The top of the post

C) A pale grub in the soil

D) The tip of the post's shadow

114. What clue leads the jay back to the original hiding place?

A) A second shadow behind the post

B) A nut lying on top of the root

C) Disturbed soil beside a band of yellow leaves

D) A grub carrying the nut toward the hole

115. What mistake causes the jay to dig in the wrong place when it first returns?

A) It assumes the nut has grown into a plant.

B) It follows the post after the post has been moved.

C) It treats the tip of a changing shadow as though it marks one fixed location.

D) It believes the root's notches have disappeared.

116. In "a root scored with two deep notches," SCORED most nearly means:

A) awarded points

B) marked with cuts or grooves

C) arranged as written music

D) judged on a rating scale

117. How does describing the shadow as a helpful "black finger" contribute to the narrative?

A) It establishes that the shadow deliberately deceives the bird.

B) It presents the narrator's unqualified guarantee that the marker cannot change.

C) It shows that the jay intends to move the cache whenever the shadow moves.

D) It expresses the jay's confidence in a pointer whose changing position readers can recognize as unreliable.

Informational Passage: Floating Without Changing the Metal

A floating vessel is pulled downward by its weight and pushed upward by the water. The upward buoyant force equals the weight of the water the vessel displaces, or pushes aside. When these upward and downward forces balance, the vessel can remain at a steady level.

A compact block of steel is heavy compared with an equal volume of water. Even when the block is fully submerged, the water it pushes aside weighs less than the block does. That available buoyant force cannot balance the block's weight.

Making a broad, hollow hull changes the relationship without changing steel into a lighter material. The vessel's outer shape encloses space that includes air. As the hull enters the water, it can push aside a large volume before being completely submerged. Its total weight can be balanced by the weight of that displaced water while part of the hull remains above the surface.

Adding cargo increases the vessel's downward weight. If the hull still has room to displace more water, it settles lower until a greater buoyant force balances the new load. The lower position does not mean the metal itself has grown heavier per unit of volume. The important quantities are the vessel's total weight and the water excluded by the submerged part of its overall shape.

118. According to the first paragraph, what is equal to the upward buoyant force?

A) The weight of all water in the harbor

B) The volume of the steel alone

C) The weight of the water displaced by the vessel

D) The weight of the vessel's air alone

119. What is the central explanation for why a hollow steel vessel can float?

A) Hollowing steel changes the density of the metal itself.

B) Water pushes upward only on objects made from lightweight materials.

C) A vessel floats because all of its metal stays above the water.

D) Its overall shape can displace water of sufficient weight to balance the vessel before the hull is fully submerged.

120. Two vessels with different hull shapes float steadily in the same water and have the same total weight. What follows from the passage?

A) They displace water of equal weight.

B) Their submerged depths must be identical.

C) They must contain equal volumes of air.

D) Their hulls must be made from the same material.

121. In "it settles lower," SETTLES most nearly means:

A) pays an outstanding bill

B) comes to rest at a lower level

C) ends an argument

D) establishes a permanent town

122. Why does the author compare a compact steel block with a hollow steel hull?

A) The contrast shows that floating depends only on which metal is chosen.

B) Keeping the material the same helps isolate the effect of overall shape and displaced volume.

C) The comparison introduces the chronological history of steelmaking.

D) The block provides a model for proving that every hollow object floats.

Humanities Passage: What a Draw Can Decide

Three households request the same unused storage room in their building. Under the rules they have agreed upon, their claims have equal priority. They write each household's name on one identical slip, fold the slips alike,

mix them in a bowl, and draw one.
The procedure can give each household an equal chance before the result is known. It cannot give all three the room. Two households will be disappointed, but their disappointment alone does not show that the mixing or drawing was biased. Equal chances and equal outcomes are different promises.
There is another question, however, that mixing cannot answer. What if one household needs temporary space to prevent serious damage to belongings, while the others want extra convenience? The residents must consider whether that difference warrants priority. Calling the claims equal must itself be warranted by reasons; putting names into a bowl does not supply those reasons.
A draw therefore has a limited role. Once an appropriate group of equal claims has been identified, chance can break the tie without pretending to discover which household is more deserving as a whole. Choosing who belongs in that group requires a different kind of judgment.
A draw can select a name; it cannot supply the reasons that put that name in the bowl.

123. How are the households represented in the initial drawing?

A) Each has one identical slip, folded alike and mixed with the others.

B) Each receives a different number of slips based on convenience.

C) The household with the largest belongings uses the largest slip.

D) Only the two households that agree to lose place slips in the bowl.

124. Which statement best captures the author's position on using a random draw?

A) A draw can resolve a tie among appropriately equal claims, but it cannot justify treating the claims as equal in the first place.

B) A draw guarantees that every participant receives the same result.

C) A draw proves that the selected household is better than the others in every respect.

D) A draw is useful only when the participants already know who will win.

125. Before using the draw in the example with unequal needs, what must the residents consider?

A) Whether the difference in need justifies assigning different priority.

B) Whether disappointment automatically makes a procedure biased.

C) Whether mixing more vigorously can determine which need is more urgent.

D) Whether chance proves that a selected household is superior in every way.

126. In "must itself be warranted by reasons," WARRANTED most nearly means:

A) paid for in advance

B) scheduled for a later time

C) left entirely to luck

D) justified

127. What does the balanced contrast in the final sentence emphasize?

A) The idea that a written name matters more than the household it represents.

B) The claim that a bowl can reason about competing needs.

C) The difference between selecting an outcome and justifying the claims that may enter the selection.

D) The requirement that the selection must occur before anyone considers eligibility.

128. How does the unequal-needs example develop the opening example of identical slips?

A) It denies that identical slips can ever be mixed fairly.

B) It qualifies the initial case by showing that equal representation is appropriate only after priority has been considered.

C) It adds instructions for making the storage room larger.

D) It shows that the first drawing must have been secretly manipulated.

Literary Passage: The Room After the Rug

The rug was the last thing we carried downstairs. Without it, our old living room answered every footstep. I said my brother's name, and the bare walls handed it back.

"Listen," I said. "It sounds wrong."

He was busy fitting the rolled rug through the doorway. "It sounds empty."

For weeks I had wanted this room to remain exactly as it was. Now the dent where the sofa had stood and a pale rectangle behind a removed picture seemed less like pieces of home than marks left by things already gone. I pressed my thumb against a dent in the wallpaper, a tiny hill I had traced while waiting for birthday guests years before.

The new tenant arrived carrying a toolbox. Beside her, a small boy held a red cushion against his chest. He looked around the room and asked where his reading chair would go.

"Perhaps by that window," she said.

He placed the cushion on the bare floor beneath it, as carefully as if the chair were already there.

My brother called from the landing. I took my hand from the wall. Before closing the door, I looked once more at the cushion. The room still echoed when I crossed it, but I no longer needed it to sound like us.

129. What does the boy place beneath the window?

A) A red cushion

B) A rolled rug

C) A small toolbox

D) A framed picture

130. Which statement best expresses the passage's central development?

A) The narrator decides that moving was a practical mistake.

B) The brother proves that an empty room has no useful purpose.

C) Seeing another family begin to imagine the room helps the narrator accept leaving it.

D) The new tenant restores the room exactly as the narrator remembers it.

131. Why does the narrator recall tracing the small dent in the wallpaper?

A) The mark proves that the new tenant damaged the room.

B) The narrator plans to hide a birthday gift inside the wall.

C) The brother has asked for evidence that the room needs repairs.

D) An ordinary physical mark carries a personal memory that makes departure difficult.

132. What does the phrase "the bare walls handed it back" do?

A) It personifies the walls to describe the echo of the narrator's voice.

B) It establishes that someone is speaking from inside the wall.

C) It compares the room's size to the narrator's hand.

D) It reports that the walls physically move toward the narrator.

133. How does the first-person point of view contribute to the ending?

A) It lets readers compare the narrator's private resistance with the narrator's later willingness to leave.

B) It reveals the new tenant's unspoken plan to replace every wall.

C) It confirms that the brother feels exactly the same attachment.

D) It explains all the boy's memories of the apartment.

Informational Passage: Why a Kernel Opens

A popcorn kernel contains a small amount of water inside a tough outer covering. Heating the kernel warms that water. As the water becomes hot vapor, pressure rises within the covering, which initially holds the contents in place. The kernel does not gradually grow to its final popped size while it remains intact.

Inside, the heated starch softens. Eventually the covering breaks because it can no longer contain the pressure. The sudden release allows the hot contents to expand rapidly. Vapor pushes through the softened starch, creating many tiny spaces. As the expanded starch cools and becomes firm, those spaces remain. This airy structure gives popped corn much of its volume.

The change depends on more than simply making a grain hot. The covering must retain pressure long enough, and the interior must contain sufficient water to produce the needed vapor. A kernel with a damaged covering can lose vapor before enough pressure builds. A very dry kernel can also fail to open properly, even though heat reaches it.

Thus the dramatic opening results from several conditions working together. Heat provides energy, water supplies vapor, the covering temporarily confines it, and softened starch forms the expanded structure after the covering gives way.

134. According to the passage, what remains in the starch as it cools and becomes firm?

A) An intact tough outer covering

B) A single pocket filled with cold water

C) Many tiny spaces created during expansion

D) The original tightly packed arrangement

135. Which statement best summarizes why popcorn opens?

A) Any dry grain expands fully as soon as its surface becomes hot.

B) Heat, internal water, a pressure-retaining covering, and softened starch cooperate to produce rapid expansion.

C) Cooling the outer covering creates vapor that draws the starch inward.

D) The covering slowly stretches until it becomes the white popped material.

136. Two otherwise similar kernels receive the same heat. One has a small crack in its covering. Why might it open less successfully?

A) The crack guarantees that no heat reaches the water.

B) The crack may let vapor escape before sufficient pressure accumulates.

C) A crack turns the starch into a substance that cannot cool.

D) The crack necessarily adds too much water to the interior.

137. In the final paragraph, "confines" most nearly means:

A) makes colder

B) keeps within limits

C) counts precisely

D) changes into light

138. Why does the author discuss damaged and very dry kernels after explaining successful opening?

A) They establish that all kernels fail for exactly the same reason.

B) They shift the subject from physical changes to farming history.

C) Their failures show why the conditions in the explanation are necessary.

D) They show that cooling alone can repair an unsuccessful kernel.

Informational Passage: A Wheel Between Pictures

A camera records a moving wheel as a series of separate pictures. Between one picture and the next, the wheel keeps moving, but those intervening positions are absent from the recording. A viewer must connect the positions that the pictures actually show.

Imagine a wheel with one bright mark on its rim. In the first picture, the mark is at the top. Before the next picture, the wheel turns three quarters of a revolution clockwise, placing the mark at the left. That second position could

also be reached by turning only one quarter of a revolution counterclockwise. The two recorded positions alone cannot reveal which journey occurred between them.

When a viewer connects each pair by the shorter apparent movement, a rapidly rotating wheel can therefore seem to turn backward. The wheel has not actually reversed. Information about its motion was lost between the recorded samples. Additional full revolutions between pictures would also go unrecorded.

Recording more pictures each second reduces the time between samples. If the wheel's movement during that interval becomes small enough, the apparent motion can agree with the actual direction. Merely enlarging the existing pictures does not supply the missing positions. The difficulty concerns when observations were made, rather than how large the recorded mark looks.

139. Where is the bright mark in the second picture of the example?

A) At the top of the wheel

B) At the bottom of the wheel

C) At the right side of the wheel

D) At the left side of the wheel

140. Which proposed change does the passage say fails to supply the missing positions?

A) Recording more pictures each second

B) Shortening the interval between pictures

C) Enlarging the existing pictures

D) Making observations at additional times

141. What central problem with recording motion does the wheel example explain?

A) Separate recorded positions can make continuous motion appear to have the wrong direction.

B) A camera forces every wheel it records to reverse direction.

C) A larger mark always produces a more accurate measurement of speed.

D) Wheels rotate only while a camera is taking a picture.

142. Suppose a wheel returns its single bright mark to exactly the same position in each picture. What can be concluded from those pictures alone?

A) The wheel must have remained still throughout the recording.

B) The wheel must have reversed direction between every pair of pictures.

C) The wheel must have moved only counterclockwise.

D) The wheel could be still or could complete whole revolutions between pictures.

143. In this passage, "samples" are:

A) small pieces removed from the wheel

B) separate observations taken at particular times

C) trial wheels made before a final model

D) colors selected for painting the rim

144. How does the final paragraph develop the earlier explanation?

A) It replaces the wheel example with an unrelated historical account.

B) It rejects the claim that any positions are missing.

C) It argues that picture size is the sole cause of apparent reversal.

D) It contrasts a change that can reduce the information gap with a change that cannot.

Literary Passage: The Coat at the Platform

Suri chose her older cousin's navy coat for her first journey alone across town. In photographs, her cousin always looked certain of where she was going. Suri hoped the coat would suggest the same certainty, although its cuffs

covered her knuckles.
At the tram stop, an attendant asked whether she knew which platform she needed.
"Of course," Suri answered before checking her ticket.
She joined the people at Platform Two. The board listed Harbor as the final stop. Her ticket said Hillrest. Perhaps, she thought, the Harbor tram went through Hillrest first. But she could not find Hillrest among the smaller names beneath it.
Her hand was closed around the ticket deep inside one sleeve. She drew it out and flattened it against the board. The names still did not match. In the window beside the board, she could see herself wearing her cousin's coat like a costume.
The attendant was still nearby.
"I think I have the wrong platform," Suri said. "Could you show me?"
"Platform Four. Good thing you checked before boarding."
The answer was so matter-of-fact that Suri wondered why asking had seemed difficult.
She folded her ticket, turned the cuffs back twice, and crossed the footbridge. The coat still looked impressive. She simply no longer expected it to know the way.

145. Which platform does the attendant tell Suri to use?

A) Platform Two

B) The Harbor platform where she is waiting

C) Platform Four

D) Any platform listed on the board

146. Which statement best captures the lesson Suri learns during the episode?

A) An impressive coat guarantees that a traveler will choose the correct route.

B) Seeking needed information is more useful than trying to appear already knowledgeable.

C) Every tram follows the same route despite different destination names.

D) Asking an attendant for help makes a journey impossible to complete.

147. Why does Suri initially imagine that the Harbor tram might pass through Hillrest?

A) The attendant has already confirmed that both stops are on the same line.

B) Her ticket instructs her to ignore all platform names.

C) She has previously traveled through Hillrest on that exact tram.

D) She is looking for a reason to keep her first choice without admitting that she needs help.

148. In the window reflection, the word "costume" suggests that Suri is:

A) preparing to act in a scheduled play

B) wearing an official tram employee uniform

C) trying to look like an experienced traveler without yet feeling like one

D) hiding the coat so nobody will recognize it

149. What is the chief effect of saying Suri no longer expects the coat "to know the way"?

A) It establishes that the coat previously supplied accurate spoken directions.

B) It proves that the attendant has hidden a map inside the coat.

C) It suggests that the coat changes the tram routes through magic.

D) The playful personification highlights her decision to rely on useful actions rather than an appearance.

Literary Passage: The Mayor's Applause

In our town, applause could be bottled. The mayor had collected enough jars to fill the shelves behind the performance hall. He called this preparation.
My job was to turn the lids when he raised one finger. A hundred captured claps would leap out, perfectly

together. When he raised two fingers, I opened the larger jar marked THUNDEROUS.
On the evening of the traveling flutist's visit, the mayor gave me an especially strict instruction: no uncertain pauses. A respectable town, he said, should always know when to admire something.
The flutist finished her first tune. Up went one finger. I released the applause. She bowed, looking toward the audience, but everyone was busy watching the mayor's hand.
During the next tune, a note rose so lightly that I forgot the shelves. The flutist let it hang, then brought it softly home. Before the mayor could signal, a child in the front row began clapping. Another pair of hands joined, then several more, out of step and unmistakably alive.
The mayor raised two fingers. I put my hand on THUNDEROUS, then took it away.
"A pause," he whispered sharply.
"No," I whispered back. "Listen."
Behind me, the unopened jars stood in their straight rows. For once, the room had no need of them.

150. What happens when the mayor raises two fingers?

A) The flutist is supposed to begin the first tune.

B) The narrator is supposed to open the jar marked THUNDEROUS.

C) The audience must leave the hall immediately.

D) The child must stop listening and close the jars.

151. Which distinction is central to the story?

A) A traveling musician always plays louder than a local musician.

B) Large jars are more useful than small ones in every performance.

C) A town can enjoy music only when all listeners clap together.

D) Organized sounds of approval differ from freely felt and expressed appreciation.

152. Why does the narrator take a hand away from the larger jar near the end?

A) Opening it would replace or cover the genuine response already developing in the room.

B) The narrator has discovered that all the jars are empty.

C) The mayor has clearly withdrawn the signal.

D) The flutist has forbidden the use of any hand movements.

153. What makes the mayor's statement that a respectable town "should always know when to admire something" ironic?

A) The narrator secretly wishes all performances would be silent.

B) His system directs attention toward his signals instead of encouraging people to respond to what deserves admiration.

C) The town has never heard any sound resembling applause.

D) The flutist openly announces that she dislikes admiration.

154. How does the opening explanation of the jars prepare readers for the ending?

A) It proves that the narrator will obey every signal without exception.

B) It supplies a scientific account of how real sound must be stored.

C) It establishes an expected routine that the narrator later chooses to interrupt.

D) It explains why the child is unable to clap.

Informational Passage: How a Zipper Joins Two Sides

A zipper can turn two separate edges into a continuous closure without glue or stitching each time it is used. Consider a zipper made of two rows of individual teeth attached to flexible strips. The teeth have matching shapes that allow a tooth on one side to fit securely between teeth on the other.
The slider does not manufacture these connections. It guides the existing teeth into the positions where their

shapes can hold together. During closing, the two separate rows enter the wider branches of a channel inside the slider. As those branches converge, the teeth are brought into alignment and pressed into engagement. A joined row leaves the narrower end.

Opening reverses the guided route. As the slider moves the other way, a dividing part inside it directs the two rows apart. The teeth separate in sequence rather than all being pulled away simultaneously. The flexible strips carry the separated rows along the opening's edges.

This mechanism depends on both the slider and the teeth. A slider can guide intact teeth, but it cannot replace a missing tooth. Likewise, teeth with suitable shapes may remain unjoined if the slider fails to bring them together properly. The useful closure results from many small connections being guided into place, not from a single bond extending along the whole opening.

155. Which component guides the two rows of teeth into alignment during closing?

A) A fresh layer of glue

B) The slider

C) A needle making new stitches

D) The fabric beyond the zipper's end

156. Which statement best explains the zipper's operation as a whole?

A) Matching teeth form many small connections as the slider guides them together or apart.

B) The slider creates new teeth every time it travels.

C) A continuous adhesive bond must be broken whenever the zipper opens.

D) The flexible strips become rigid before any teeth can connect.

157. Why might moving an undamaged slider repeatedly fail to repair a gap caused by a missing tooth?

A) Repeated movement always removes every remaining tooth.

B) The strips must stop being flexible before the slider works.

C) The slider can connect the edges only by adding glue.

D) Guiding the rows cannot supply the connecting structure that is absent.

158. In the phrase "pressed into engagement," "engagement" most nearly means:

A) a fitted connection that holds the parts together

B) a promise to attend an event

C) a period of paid employment

D) an argument between opposing groups

159. What process is explained immediately after the paragraph describing closing?

A) Opening the zipper by guiding its rows apart

B) Manufacturing the flexible strips

C) Adding a second row of stitches

D) Choosing a color for the teeth

Literary Passage: A Place to Stop

I slipped my sketchbook into the bicycle bag, though I expected our ride to leave no time for drawing. Uncle Pavo had taught me to ride along this harbor road. In those days, whenever I fell behind, his chain seemed to require a mysterious adjustment. By the time he finished inspecting it, I had caught up.

Now I reached the top of the rise first. When I looked back, his pedaling had become labored. He waved me onward, but the space between our bicycles kept widening.

Beside the old loading dock, I stopped and opened my sketchbook.

"That crane," I called when he arrived. "I want to draw it."

He leaned his bicycle against the railing and rested his forearms beside mine. I made three quick lines on the

page. He glanced at them, then at the crane's web of cables.
"Plenty of ropes to get right," he said. "Take your time."
I wondered whether he remembered all those troublesome chains. Neither of us mentioned them. Below us, a small boat worked its way past the dock. Uncle Pavo pointed out the reflection of its red hull, and I added a streak beneath my crane.
The harbor was taking longer to draw than I had expected. I did not mind. I began a second page.

160. What does the narrator say is the reason for stopping beside the loading dock?

A) Needing to adjust a broken chain

B) Waiting for a ticket seller

C) Wanting to draw the crane

D) Searching for a missing bicycle bag

161. What does Uncle Pavo's remark about the many ropes most strongly suggest?

A) He has decided the narrator must become a professional dock worker.

B) He believes a bicycle chain is attached to the crane.

C) He accepts a reason for making the stop last longer without discussing his need to rest.

D) He wants the narrator to finish the drawing immediately.

162. In "his pedaling had become labored," "labored" most nearly means:

A) paid for by the hour

B) done with noticeable effort

C) performed without movement

D) completed before the ride began

163. How does the recollection of the uncle's earlier chain adjustments contribute to the present scene?

A) It proves that both bicycles still have identical mechanical faults.

B) It creates a parallel in which the person quietly making room for the other's slower pace has changed.

C) It explains why the narrator has never enjoyed the harbor.

D) It interrupts the story with an unrelated account of bicycle manufacture.

164. What is the effect of leaving any direct discussion of the uncle's fatigue out of the scene?

A) Their consideration for each other emerges through actions and hints instead of an explicit announcement.

B) Readers are told that the two characters cannot communicate at all.

C) The narrator becomes unaware that the bicycles have separated.

D) The passage changes into an impersonal set of riding instructions.

Informational Passage: Why Some Gloves Work on a Screen

Many touchscreens respond to an electrical effect rather than to pressure alone. In a simplified description, a capacitive screen maintains an electric field near its surface. A conductive object, such as a bare finger, can alter that field locally. The screen's electronics detect the change and estimate where contact occurred. Here, conductive means able to allow electric charge to move.
An ordinary insulating glove can interfere with this process. Its material separates the finger from the screen and may prevent a strong enough electrical interaction. Pressing harder does not necessarily solve the difficulty: additional pressure does not turn an insulating material into a conductor.
Some gloves contain conductive threads in their fingertips. When their design provides a suitable electrical connection, those threads let the covered finger interact with the screen. Their usefulness does not come merely from being soft, thin, or easy to bend. Two gloves that feel equally flexible can behave differently if their electrical properties differ.
This account does not describe every touchscreen. Some designs detect pressure through a different

mechanism, so a glove that fails on one type may work on another. The useful question is therefore not simply whether a glove can touch the glass. It is whether that contact produces the particular kind of signal the screen is designed to detect.

165. What do the electronics in the described capacitive screen detect?

A) The color of the glove

B) A local change in an electric field

C) Only the sound of a fingertip touching glass

D) The user's spoken instructions

166. Which statement best expresses the passage's main explanation?

A) All screens work by measuring how hard a finger presses.

B) Every thin glove works on every kind of touchscreen.

C) A glove must allow the kind of interaction a screen detects, so contact or flexibility alone may not be enough.

D) Conductive threads are useful only because they make fabric softer.

167. Two gloves are equally thin and flexible, but only the one with conductive fingertips works on a particular screen. Which explanation is best supported?

A) The successful glove must press much harder because both are equally flexible.

B) The unsuccessful glove proves that the screen cannot detect a bare finger.

C) Its electrical properties provide an interaction that the other glove does not provide.

D) The successful glove changes the screen into a pressure-only device.

168. In "a suitable electrical connection," "suitable" most nearly means:

A) appropriate for the required purpose

B) visible from a distance

C) identical in color

D) unusual in appearance

169. Why does the final paragraph mention screens that detect pressure through another mechanism?

A) To claim that the earlier electrical explanation is false for all screens.

B) To establish that every glove has identical electrical properties.

C) To change the subject to the manufacture of ordinary windows.

D) To limit the explanation's scope and clarify why results can differ between screen types.

Humanities Passage: The Meals Missing from a Ledger

Imagine finding an old innkeeper's account book. Its pages list purchases of flour, candles, coal, and vegetables. The entries offer valuable evidence of the inn's daily operation, but they do not provide a complete record of everything used there. The book was made to track payments, not to describe every activity.

Water drawn from a well might leave no purchase entry. Vegetables from a nearby household's field could arrive as a gift. An assistant might carry out a task without being paid separately for that occasion. None of these possibilities proves what actually happened at the inn. They do show why silence in an expense record need not mean absence from daily life.

The distinction matters when comparing periods. A year with more vegetable purchases might reflect greater consumption, but it might also reflect a change from gifts to paid supplies. Before choosing an explanation, a historian could look for letters, instructions, or other records that clarify the circumstances.

An account book is not useless because it is incomplete for these broader questions. Its strength lies in the transactions it was designed to record. Careful interpretation begins by asking what the keeper needed to write down, then distinguishing that purpose from the wider story a later reader hopes to reconstruct.

170. For what original purpose was the imagined account book created?

A) To track payments

B) To describe every meal in complete detail

C) To record all conversations at the inn

D) To list only unpaid gifts

171. Which principle best represents the author's central point?

A) A record that omits any activity has no historical value.

B) A missing purchase entry proves that an item was never used.

C) Later readers should assume every account book describes the same circumstances.

D) A historical record should be interpreted in light of what it was created to document.

172. Suppose vegetable-purchase entries increase sharply in one year. Which further finding would support an explanation other than increased consumption?

A) An unrelated page lists the color of the inn's shutters.

B) A letter states that a neighbor stopped giving the inn vegetables that year.

C) A candle purchase appears in the same account book.

D) The book contains ordinary entries written on lined paper.

173. In "the wider story a later reader hopes to reconstruct," "reconstruct" most nearly means:

A) pay for in advance

B) copy every word without interpretation

C) replace with a more entertaining invention

D) piece together from available evidence

174. How does the last paragraph respond to the limitations discussed earlier?

A) It abandons the ledger in favor of guessing what happened.

B) It asserts that incomplete records must always be discarded.

C) It denies that the keeper had any reason for recording payments.

D) It preserves the ledger's value while defining the kinds of conclusions it can support most directly.

Section 4: Mathematics

64 questions | Recommended time: 45 minutes

175. Factorials satisfy n! = n × (n − 1)! for positive integers n. Since 1! = 1, what value of 0! makes the rule work when n = 1?

A) 1

B) 0

C) −1

D) 10

176. A fraction a/b in lowest terms has a positive odd denominator b. Add 1/2 and reduce the result to lowest terms. What must be true of the new denominator?

A) It is odd.

B) It is even.

C) It is always 1.

D) It must equal b.

177. The number of positive divisors of an integer is found by adding 1 to every exponent in its prime factorization and multiplying those results. What is the least positive integer with exactly 12 positive divisors?

A) 48

B) 60

C) 72

D) 96

178. What is the absolute value of 2 minus the absolute value of (3 − 8)?

A) −3

B) 5

C) 7

D) 3

179. An expression linear in x has the form mx + b, where m and b are constants. Treat a as a fixed nonzero constant. Which expression is linear in x?

A) a/x

B) x^2/a

C) x/a

D) a/x^2

180. A linear function has the rule f(x) = ax + b. If f(2) > f(5), which statement must be true?

A) a > 0

B) a = 0

C) a < 0

D) b = 0

181. A function maps integers to integers and is strictly increasing: whenever u < v, f(u) < f(v). If f(1) = 2 and f(4) = 5, what is f(3)?

A) 3

B) 5

C) It cannot be determined.

D) 4

182. A marker's position p, in meters, is given by p = 6t + 5 when t is measured in minutes. If s measures the same elapsed time in seconds, which formula gives p in terms of s?

A) p = s/10 + 5

B) p = 360s + 5

C) p = 6s + 5

D) p = s/10 + 300

183. The diagonals of parallelogram ABCD meet at O. If AO = 7 centimeters, what is OC?

A) 5 centimeters

B) 14 centimeters

C) 7 centimeters

D) 21 centimeters

184. A square pyramid has a base with side length 12 units. A plane parallel to the base cuts the pyramid halfway between the vertex and the base. What is the cross-section's area?

A) 72 square units

B) 36 square units

C) 144 square units

D) 24 square units

185. Two congruent equilateral triangles share the same center. One is rotated 180° about that center relative to the other, forming a six-pointed star. Each triangle has area 81 square units. What is the area common to both triangles?

A) 27 square units

B) 40.5 square units

C) 162 square units

D) 54 square units

186. A complete musical excerpt has 24 measures of 3 beats each. It is played at a steady 96 beats per minute, with no extra beats or pauses. How long does it last?

A) 24 seconds

B) 72 seconds

C) 90 seconds

D) 45 seconds

187. A shape covers 9 grid squares completely and 6 more partially, with no other covered squares. Each square has area 1 square centimeter. Every partially covered square contributes strictly between 0 and 1 square centimeter. Let A be the numerical value of the total area in square centimeters. Which statement must be true?

A) $9 < A < 12$

B) $9 < A < 15$

C) $12 < A < 15$

D) $A = 12$

188. Volunteers have already enrolled in a study. A coin flip determines which of two packing methods each volunteer will try. What does the coin flip perform?

A) Random sampling of the entire population

B) A census of all potential volunteers

C) Random assignment to treatment groups

D) A measurement of each volunteer's packing speed

189. Exactly 2% of a batch of 1,000 parts are cracked. An inspection flags 90% of the cracked parts and 5% of the uncracked parts. A part is selected uniformly from the flagged parts. What is the probability that it is cracked?

A) 18/67

B) 9/10

C) 1/50

D) 18/1000

190. Four lamps independently show amber or blue, each with equal probability. A color change is counted whenever neighboring lamps differ. What is the probability of exactly two changes across the three neighboring pairs?

A) 3/8

B) 3/16

C) 1/4

D) 1/2

191. A design has 25 equal squares. If 36% of the squares are colored, how many squares remain uncolored?

A) 9

B) 11

C) 16

D) 21

192. An alloy combines 3 kilograms of material A with 2 kilograms of material B. Copper makes up 1/6 of A and 1/4 of B by mass. What fraction of the finished alloy is copper?

A) 1/12

B) 1/5

C) 5/24

D) 5/12

193. Neither fraction in any listed sum has a terminating decimal. Which sum nevertheless has a terminating decimal?

A) 1/6 + 1/3

B) 1/3 + 1/9

C) 1/6 + 1/9

D) 1/3 + 1/12

194. A recording booth charges C = 9 + 4h dollars for h hours of use. What does the 9 represent?

A) The hourly charge

B) The maximum permitted hours

C) The cost of four hours of use

D) A fixed charge independent of the number of hours

195. A flexible border is assembled from x strips, each 6 centimeters long, in a single chain. Each joint overlaps 2 centimeters, and no other overlaps occur. Which expression gives the finished length?

A) 4x centimeters

B) 4x + 2 centimeters

C) 6x − 2 centimeters

D) 8x − 2 centimeters

196. What value of x makes $(x - 4)^2 = (x + 2)^2$ true?

A) −1

B) 2

C) 3

D) 1

197. A machine halves an even integer and adds 5 to an odd integer. Its output is fed through the same machine once more. Which set lists ALL positive integer starting values that produce a final output of 6?

A) 2, 7, and 24

B) 2 and 24 only

C) 7 and 12 only

D) 1, 6, and 12

198. Point P is at (−3, 5). It is reflected across the x-axis and then across the y-axis. What are its final coordinates?

A) (3, −5)

B) (−3, −5)

C) (3, 5)

D) (−5, 3)

199. The parallel bases of a trapezoid are 8 and 16 centimeters long. What is the length of the segment joining the midpoints of its nonparallel sides?

A) 4 centimeters

B) 8 centimeters

C) 24 centimeters

D) 12 centimeters

200. A 10-by-8 rectangular panel has all four corners trimmed. Each removed corner is a right triangle with both legs 2 units long, lying along the original edges. What is the new perimeter?

A) 20 + 4√2 units

B) 20 + 8√2 units

C) 28 + 8√2 units

D) 36 − 8√2 units

201. One tablespoon equals 3 teaspoons, and one teaspoon holds 5 milliliters. How many milliliters are in 2 tablespoons plus 1 teaspoon?

A) 20 milliliters

B) 30 milliliters

C) 35 milliliters

D) 45 milliliters

202. A batch of fruit weighs 1.8 kilograms before peeling. Peeling removes 18% of its mass. How many complete 82-gram portions can be made from the remaining fruit?

A) 17

B) 20

C) 18

D) 21

203. A flight leaves at 22:40 Monday from a place using UTC−3 and lasts 3 hours 35 minutes. The destination uses UTC+2. What is the local arrival time? Assume neither place changes its clock.

A) 02:15 Tuesday

B) 05:15 Tuesday

C) 07:15 Monday

D) 07:15 Tuesday

204. Six measured lengths are 7, 7, 8, 11, 17, and 20 centimeters. How much greater is their range than their mode?

A) 7 centimeters

B) 6 centimeters

C) 9.5 centimeters

D) 13 centimeters

205. Three different cards are chosen at random from cards numbered 1 through 6. Every set of three cards is equally likely. What is the probability that their sum is divisible by 3?

A) 1/5

B) 1/3

C) 2/5

D) 3/5

206. The ordered data are 5, 8, 9, 12, 14, 18, and 21. The smallest and largest values are removed. Which statistic keeps the same value?

A) Median

B) Mean

C) Range

D) Number of observations

207. A sample has mass 15/8 gram. Between which pair of consecutive half-gram marks would it lie on a scale?

A) 1 gram and 1.5 grams

B) 2 grams and 2.5 grams

C) 2.5 grams and 3 grams

D) 1.5 grams and 2 grams

208. Two invoices begin with the same item price. Invoice A discounts the item by 25%, then adds a $5 handling fee. Invoice B adds the fee first, then discounts the entire amount by 25%. How much more does A charge?

A) $0

B) $1.25

C) $3.75

D) $5

209. Which fraction is greater than −5/6 but less than −3/4?

A) −7/8

B) −17/24

C) −11/12

D) −19/24

210. The interval from −1/2 to 5/3 is divided into seven equal parts. What number is at the fourth division point to the right of −1/2?

A) 31/42

B) 13/42

C) 5/7

D) 16/21

211. For which value of x is the expression (x + 5)/(x − 3) undefined?

A) −5

B) −3

C) 3

D) 5

212. The numbers a and b satisfy 2a + 3b = 11. What is the value of 6a + 9b − 4?

A) 7

B) 29

C) 18

D) 33

213. A conveyor can support 50 kilograms, but the operator keeps a 3-kilogram safety margin. A 12-kilogram load is already on it. At most how many additional 7.5-kilogram parcels can be added?

A) 4

B) 3

C) 5

D) 6

214. A full box contains n cards. Removing 3 cards leaves at least half the original number; removing 7 leaves less than one third. Each removal starts from the full box, and both are possible. How many positive integer values can n have?

A) 3

B) 5

C) 4

D) 6

215. A square turns 90 degrees about its center. Which point stays in exactly the same location throughout the turn?

A) A vertex

B) The midpoint of one side

C) A point halfway between the center and a vertex

D) The center

216. A right triangular prism has triangular bases with legs 3 and 4 centimeters. The prism is 7 centimeters long. What is the area of its largest rectangular face?

A) 21 square centimeters

B) 35 square centimeters

C) 28 square centimeters

D) 42 square centimeters

217. Triangle ABC has area 96 square units. Point D is the midpoint of AB. Point E lies on AC, with AE equal to one third of AC. What is the area of triangle BDE?

A) 16 square units

B) 32 square units

C) 48 square units

D) 80 square units

218. A water meter increases from 2.750 cubic meters to 3.125 cubic meters. One cubic meter equals 1,000 liters. How many liters were used?

A) 37.5 liters

B) 3,125 liters

C) 375 liters

D) 5,875 liters

219. A plan drawn at a scale of 1:200 is enlarged to 125% of its original linear size. A wall measures 12.5 centimeters on the enlarged copy. How long is the actual wall?

A) 16 meters

B) 20 meters

C) 25 meters

D) 31.25 meters

220. A trail survey is answered only by users who choose to scan a posted code. Why might its results fail to represent all trail users?

A) Scanning a code guarantees that every user responds.

B) Voluntary responses always give equal representation to every opinion.

C) People who choose to respond may differ from those who do not.

D) A large number of responses automatically eliminates selection bias.

221. Two different keys numbered 1 through 4 are selected and placed left and right, with every ordered pair equally likely. Given that the left number exceeds the right, what is the probability that the left number is even?

A) 1/3

B) 1/2

C) 3/4

D) 2/3

222. A survey divides all respondents into three nonoverlapping categories. Their displayed percentages, rounded individually to whole numbers, are 34%, 34%, and 33%. Which interpretation is justified?

A) Individual rounding can explain the displayed total of 101% without an error in the underlying counts.

B) The categories must overlap despite the description.

C) Every percentage must have been calculated incorrectly.

D) The three categories must contain equal numbers of respondents.

223. A first batch contains 5 defective parts out of 20. A second contains 8 out of 40. Which statement correctly compares the defect rates?

A) The second rate is higher because 8 is greater than 5.

B) The rates are equal because both batches contain defects.

C) The second rate is lower by 20 percentage points.

D) The second rate is lower by 5 percentage points.

224. The temperature rises from −6°C at dawn to 10°C at noon. By evening it has fallen by three fourths of that morning increase. What is the evening temperature?

A) −2°C

B) 2°C

C) 4°C

D) 7.5°C

225. A number is the product of ten factors of 2 and seven factors of 5. How many digits does it have in ordinary base-ten notation?

A) 7

B) 9

C) 17

D) 8

226. Which expression is negative for EVERY real number $x > 0$?

A) $x - 1$

B) $-(x + 1)$

C) x^2

D) $1/x$

227. For $x \neq 2$, solve $x/(x - 2) = 3$.

A) −3

B) 2

C) 3

D) 6

228. The function g is defined by g(x) = 2x − 5. If g(g(t)) = 7, what is t?

A) 6

B) 11/2

C) 11

D) 22

229. A program contains only 3-minute and 5-minute clips, with no gaps, and lasts exactly 31 minutes. What is the smallest possible number of clips?

A) 6

B) 9

C) 11

D) 7

230. A triangle is reflected across a vertical line. Which quantity is ALWAYS preserved?

A) Every vertex's x-coordinate

B) The left-to-right order of all its vertices

C) Its exact position on the page

D) Its area

231. A circular disc has radius 10 centimeters. A concentric circular hole is removed so that the remaining ring has 75% of the original area. What is the hole's radius?

A) 2.5 centimeters

B) 7.5 centimeters

C) 5 centimeters

D) 5√3 centimeters

232. A cube is built from 27 unit cubes in a 3-by-3-by-3 arrangement. The unit cube at the center of its top face is removed. What is the total area of all exposed surfaces, including the new cavity?

A) 58 square units

B) 53 square units

C) 54 square units

D) 60 square units

233. A film begins at 17:48 and ends at 19:15. That interval includes a 10-minute intermission. How many minutes of film are shown?

A) 67 minutes

B) 87 minutes

C) 77 minutes

D) 97 minutes

234. A pump delivers 12 liters per minute while running. After every 90 seconds of pumping, it pauses for 30 seconds before running again. Starting with an empty tank, how long does it take to deliver 36 liters?

A) 3 minutes

B) 4 minutes

C) 3 minutes 30 seconds

D) 4 minutes 30 seconds

235. A fabric strip is 150 centimeters wide and 2.4 meters long. Its mass is 250 grams per square meter. What is the strip's mass?

A) 90 grams

B) 900 grams

C) 600 grams

D) 9,000 grams

236. A log records 0 delays on 6 days, 1 delay on 3 days, and 2 delays on 1 day. How many days are recorded altogether?

A) 5

B) 10

C) 6

D) 9

237. A fair six-sided die awards 0 points on 1, 2, or 3; 2 points on 4 or 5; and k points on 6. What value of k makes the theoretical average award 2 points per roll?

A) 8

B) 2

C) 4

D) 12

238. Ten readings are grouped as follows: 2 readings from 0 through 9, 5 from 10 through 19, and 3 from 20 through 29. Which interval must contain the median?

A) 10 through 19

B) 0 through 9

C) 20 through 29

D) No interval can be identified without knowing every reading.

Section 5: Language

60 questions | Recommended time: 30 minutes

239. Which sentence places a matched pair of parentheses around ONLY the item label a?

A) Choose item (a) from the list.

B) Choose item a) from the list.

C) Choose item (a from the list).

D) Choose item (a from the list.

240. These original lines are quoted within one line of prose. Which version uses the standard mark for the line break?
The lamps go out
the harbor waits

A) "The lamps go out, the harbor waits"

B) "The lamps go out / the harbor waits"

C) "The lamps go out ... the harbor waits"

D) "The lamps go out: the harbor waits"

241. Which reference uses standard punctuation to mean EVERY page from 18 through 24?

A) pages 18:24

B) pages 18;24

C) pages 18–24

D) pages 18, 24

242. A company styles its name eBay. Which sentence preserves that capitalization within a sentence?

A) I found the listing on Ebay.

B) I found the listing on eBay.

C) I found the listing on EBay.

D) I found the listing on ebay.

243. Use headline-style capitalization, including the first word of a subtitle. Which form is correct?

A) Night Routes: a Guide to the Back Roads

B) Night Routes: A Guide To The Back Roads

C) Night routes: A guide to the back roads

D) Night Routes: A Guide to the Back Roads

244. Which spelling correctly names a small baked bread or pastry?

A) bisquit

B) biscuit

C) biscut

D) biscuitte

245. Which spelling correctly names an individual's distinctive personal peculiarity?

A) idiosyncracy

B) ideosyncrasy

C) idiosyncrasy

D) idiosincrasy

246. Choose the pair that correctly completes the sentence.
Several unusual ___ were recorded, but only one ___ was explained.

A) phenomenon; phenomena

B) phenomena; phenomena

C) phenomena; phenomenon

D) phenomenon; phenomenon

247. Choose the word that correctly begins this inverted conditional sentence.
___ the projector fail, we can use the printed charts instead.

A) Would

B) Had

C) Were

D) Should

248. Choose the standard verb phrase that states Zora's current knowledge.
Zora ___ the combination to her own suitcase.

A) is knowing

B) know

C) has knew

D) knows

249. Choose the pair that correctly completes the sentence in formal English.
The grant may ___ the expense, while the long distance could ___ against participation.

A) mitigate; militate

B) militate; mitigate

C) mitigate; mitigate

D) militate; militate

250. The ceremony requires a particular opening and forbids advertising. Choose the pair that correctly completes the sentence.
Its rules ___ a specific opening formula and ___ promotional messages.

A) prescribe; proscribe

B) proscribe; prescribe

C) prescribe; prescribe

D) proscribe; proscribe

Editing Passage: Keeping a Few Old Words

(1) A learner's dictionary can help readers by retaining a selected number of obsolete words. (2) An obsolete word has passed out of ordinary current use, but it may still appear in an older letter or poem. (3) A marked entry can help a reader understand that text. (4) Recording the word does not require recommending it for a new email. (5) Space is limited, so current words must remain easy to find. (6) For that reason, every obsolete word deserves the same space as a common current one. (7) Selection should depend on how useful an older entry is likely to be for the dictionary's readers. (8) The goal is to support understanding of earlier language while keeping

the guide useful for present-day communication.

251. Which title best represents the passage's limited proposal?

A) Why Every Obsolete Word Should Return to Daily Speech

B) How to Replace Modern Words with Older Ones

C) Why Dictionary Space Never Runs Out

D) Why Selected Older Words Still Belong in a Dictionary

252. Which sentence added after sentence 5 would concede the concern about space while preserving the main proposal?

A) A learner's dictionary should give current words most of its space, even if it retains a few older ones.

B) Limited space means that every older word must be excluded.

C) Older words should replace all current entries to make more room.

D) An entry's usefulness to readers need never affect its inclusion.

253. Which revision of sentence 6 best removes its overstatement while maintaining the passage's purpose?

A) Every word, however rarely encountered, should receive an equally long entry.

B) Only words that everyone uses daily can help any reader.

C) A limited selection of older entries can be useful when their value to readers justifies the space.

D) A dictionary can recommend older words only by removing all modern ones.

254. Which sentence correctly punctuates a direct question quoted within a statement?

A) The guide asked, "Which path reaches the saddle first"?

B) The guide asked? "Which path reaches the saddle first."

C) The guide asked, "Which path reaches the saddle first."?

D) The guide asked, "Which path reaches the saddle first?"

255. Which version correctly punctuates one quoted sentence interrupted by the speaker's identification?

A) "The ink." the engraver said, "will dry unevenly."

B) "The ink," the engraver said, "will dry unevenly."

C) "The ink," the engraver said "will dry unevenly."

D) "The ink," the engraver said; "will dry unevenly."

256. Which sentence correctly distinguishes the planet's name from a common word for soil?

A) From earth, the probe carried a small sealed sample of Earth.

B) From Earth, the probe carried a small sealed sample of earth.

C) From Earth, the probe carried a small sealed sample of Earth.

D) From earth, the probe carried a small sealed sample of earth.

257. Which sentence correctly capitalizes a US region while leaving a compass direction lowercase?

A) They traveled south to study traditional crafts in the South.

B) They traveled South to study traditional crafts in the south.

C) They traveled South to study traditional crafts in the South.

D) They traveled south to study traditional crafts in the south.

258. Choose the correct spelling of the singular noun for a plant with thick, water-storing stems.

A) caktus

B) cactuss

C) cactis

D) cactus

259. Choose the correctly spelled verb: "The craftsperson will ___ the blade before carrying it."

A) sheeth

B) sheathe

C) sheatth

D) shethe

260. Which spelling correctly completes the sentence? "The joyful music was sure to ___ the listeners."

A) exhilerate

B) exhiliarate

C) exhillarate

D) exhilarate

261. Choose the form that correctly completes the sentence: "The technicians let the small glider ___ freely along the wire."

A) to drift

B) drift

C) drifting

D) drifts

262. Choose the completion that avoids a misplaced modifier: "Suspended from four cables, ___."

A) the operators watched the platform carefully

B) the motor controlled the height of the platform

C) the platform remained level above the water

D) the inspector checked the platform from the pier

263. Which sentence correctly reports the question "Why did the clock stop?" after "Nobody knows"?

A) Nobody knows why the clock stopped.

B) Nobody knows why did the clock stop.

C) Nobody knows why the clock did stopped.

D) Nobody knows why stopped the clock.

264. Which sentence uses DISINTERESTED specifically to mean impartial, rather than bored or inattentive?

A) Disinterested in the dull broadcast, Arlo switched off the radio.

B) The disinterested audience stopped listening and began to yawn.

C) With no stake in either side's success, the disinterested referee considered both claims carefully.

D) Too disinterested to finish the puzzle, Mina left it untouched.

265. An editor removes restricted details from a report. A committee formally condemns an official's conduct. Which pair correctly describes these different actions? "The editor will ___ the report; the committee will ___ the official."

A) censure; censor

B) censor; censor

C) censor; censure

D) censure; censure

Editing Passage: A Sound Check That Finds Trouble

(1) A sound check should test the equipment in conditions close to those of the coming performance. (2) First, the operator connects each microphone to the channel that will carry it during the event. (3) The performer then speaks from the position that will actually be used. (4) A test from beside the control desk may miss problems caused by the performer's distance from a speaker. (5) The operator listens for unwanted noise. (6) The operator also checks whether ordinary speech is clear. (7) Many concert posters use bright colors to attract attention. (8) After adjustments, the performer repeats the test without changing position. (9) Changing several conditions at once can make it difficult to identify what corrected a problem.

266. Which sentence should be removed because it does not contribute to the passage's focus?

A) Sentence 2

B) Sentence 4

C) Sentence 8

D) Sentence 7

267. Which revision best combines sentences 5 and 6 while preserving both checks?

A) The operator listens for unwanted noise and checks whether ordinary speech is clear.

B) Because speech is always clear, the operator listens only for unwanted noise.

C) The operator listens for noise instead of checking ordinary speech.

D) If unwanted noise stops, the operator no longer needs to test speech.

268. Where should this sentence be inserted? "Keeping the same position makes the before-and-after comparison more useful."

A) After sentence 8

B) After sentence 1

C) After sentence 2

D) After sentence 5

269. Two pilots each have a flight bag. Which phrase correctly shows that the bags belong to those pilots?

A) the pilots' flight bags

B) the pilot's flight bags

C) the pilots flight bag's

D) the pilots' flight bag's

270. Which version correctly joins the independent clauses with the transition however?

A) The channel was narrow, however, the barge passed safely.

B) The channel was narrow; however the barge passed safely.

C) The channel was narrow however; the barge passed safely.

D) The channel was narrow; however, the barge passed safely.

271. Which sentence most clearly separates three destinations when each city is followed by its state?

A) The route includes Dover; Delaware, Salem; Oregon, and Helena; Montana.

B) The route includes Dover, Delaware; Salem; Oregon, and Helena, Montana.

C) The route includes Dover, Delaware; Salem, Oregon; and Helena, Montana.

D) The route includes Dover Delaware; Salem Oregon; and Helena Montana.

272. Which sentence correctly capitalizes the name of a language?

A) The crew spoke italian while crossing the canal.

B) The Crew spoke italian while crossing the canal.

C) The crew Spoke italian while crossing the canal.

D) The crew spoke Italian while crossing the canal.

273. Which phrase correctly distinguishes a named historical period from a century description?

A) renaissance art from the Fifteenth Century

B) Renaissance art from the fifteenth century

C) Renaissance art from the Fifteenth century

D) renaissance Art from the fifteenth Century

274. Choose the correct spelling for the musical instrument played by expanding and compressing bellows.

A) accordian

B) accordion

C) acordion

D) accordeon

275. Which spelling correctly completes the sentence? "A period of one thousand years is a ___."

A) millennium

B) millenium

C) milennium

D) milleniumm

276. Choose the grammatically correct comparison: "With the new rollers, the drawer moves ___ than before."

A) more smoother

B) most smoothly

C) more smoothly

D) smoothliest

277. Choose the verb that agrees with the subject: "All but one of the channels ___ clear."

A) are

B) is

C) was

D) has been

278. Choose the correct pronoun: "The award should go to ___ the panel believes has made the strongest contribution."

A) whomever

B) whose

C) whoever

D) whom

279. Choose the word meaning an upward climb: "The final ___ to the ridge took an hour."

A) assent

B) accent

C) ascent

D) access

280. Choose the pair that correctly completes the sentence: "The earlier decision established a ___; in the new plan, safety takes ___ over the timetable."

A) precedent; precedence

B) precedence; precedent

C) precedent; precedent

D) precedence; precedence

281. The path has many bends, and the prolonged noise is painfully unpleasant. Which pair states these meanings precisely? "The path is ___; the noise is ___."

A) torturous; tortuous

B) tortuous; tortuous

C) torturous; torturous

D) tortuous; torturous

Editing Passage: Two Routes Through a Page

(1) A footnote can provide a useful second path through an explanation. (2) The main text presents the central idea, while a note may supply the origin of a term or an extra example. (3) Readers who need only the main explanation can continue without stopping. (4) Curious readers can follow the note and then return to the same place. (5) This arrangement offers a choice about how much detail to explore. (6) ___, an essential step should not appear only in a footnote. (7) If skipping that step makes the main explanation impossible to follow, the note no longer provides an optional path. (8) The writer should move that necessary information into the main text.

282. Which transition best fills the blank in sentence 6?

A) Similarly

B) However

C) For example

D) As a result

283. Which sentence would best conclude this passage?

A) Readers should always skip every note they encounter.

B) A useful footnote adds depth while leaving a complete route through the main explanation.

C) All examples and definitions belong outside the main text.

D) A page becomes clearer whenever more of its explanation is hidden below it.

284. Which sentence correctly punctuates the introductory phrase and the list of colors?

A) To create, the tint mix red white, and a touch of blue.

B) To create the tint mix, red, white, and a touch of blue.

C) To create the tint, mix red, white, and a touch of blue.

D) To create the tint, mix red white and, a touch of blue.

285. Which sentence correctly uses a colon to introduce a list?

A) The stencil requires: a knife, a mat, and a ruler.

B) The three required tools are: a knife, a mat, and a ruler.

C) The stencil requires three tools: a knife, a mat, and a ruler.

D) For this stencil, use: a knife, a mat, and a ruler.

286. Which sentence correctly capitalizes the holiday and day names?

A) The ferry runs on labor day but not every Monday.

B) The ferry runs on Labor day but not every monday.

C) The Ferry runs on labor Day but not every monday.

D) The ferry runs on Labor Day but not every Monday.

287. Which version correctly capitalizes the geographic names?

A) They crossed the strait of Gibraltar before entering the Mediterranean sea.

B) They crossed the Strait of Gibraltar before entering the Mediterranean Sea.

C) They crossed the Strait Of Gibraltar before entering the Mediterranean Sea.

D) They crossed the Strait of gibraltar before entering the mediterranean Sea.

288. Choose the correctly spelled word for an open-sided roofed structure used for shelter or gatherings.

A) pavillion

B) pavilan

C) pavilon

D) pavilion

289. Which spelling correctly completes the sentence? "The confectioner sprinkled a chopped ___ over the pastry."

A) pistacio

B) pistachio

C) pistachoe

D) pistachiao

290. Which word uses the standard US English spelling?

A) zucchini

B) zuchini

C) zuccini

D) zucchinni

291. Choose the correct verb form: "By morning, the last leaf had ___ from the branch."

A) fell

B) fall

C) falling

D) fallen

292. Choose the verb that agrees with the subject: "Two thirds of the cement ___ still wet."

A) are

B) is

C) were

D) be

293. Which sentence uses the correct word order after the limiting phrase Only after?

A) Only after the bridge opened traffic did resume.

B) Only after the bridge opened did resume traffic.

C) Only after the bridge opened did traffic resume.

D) Only after the bridge opened traffic resume did.

294. Choose the pair that correctly completes the sentence: "The magazine published the story as a ___; breakfast included ___ and milk."

A) serial; cereal

B) cereal; serial

C) serial; serial

D) cereal; cereal

295. Choose the correct pair: "The farm sells its milk through a ___; the traveler records each day's events in a ___."

A) diary; dairy

B) dairy; dairy

C) diary; diary

D) dairy; diary

296. Choose the pair that completes the sentence precisely: "The exhibition covers the full ___ of styles; the unexpected opening offer was a negotiating ___."

A) gamut; gambit

B) gambit; gamut

C) gamut; gamut

D) gambit; gambit

Editing Passage: A Return with a Difference

(1) [Topic sentence goes here.] (2) That later mark gains meaning because readers can compare it with the untouched material shown earlier. (3) For example, a story opens with an untouched lump of clay resting on a pottery wheel. (4) At the end, a ring of dried clay remains on the wheel, but the lump has gone. (5) Readers can infer that the character has begun working, even if a completed pot is never described. (6) The return to a familiar object gives the ending a connection to the opening. (7) Repeating the first image without any change, however, might suggest that nothing has happened. (8) The writer should therefore consider what the altered image lets readers understand.

297. Which topic sentence best belongs in sentence 1?

A) A pottery wheel should always be cleaned with a dry cloth.

B) Readers should ignore the opening once a story has ended.

C) Returning to a familiar image in a changed state can give a story's ending a sense of completion.

D) Every completed story must describe a finished pot.

298. Which order of sentences 2, 3, and 4 most clearly introduces the example, shows its change, and then explains that change's significance?

A) 3, 4, 2

B) 2, 4, 3

C) 4, 2, 3

D) 3, 2, 4

Answer Key

Question numbers continue across all five sections.

Verbal Skills

1. C	2. A	3. D	4. D	5. C	6. D	7. B	8. B
9. B	10. C	11. A	12. A	13. B	14. A	15. C	16. D
17. D	18. A	19. B	20. D	21. A	22. C	23. B	24. D
25. B	26. B	27. C	28. C	29. C	30. A	31. B	32. C
33. D	34. D	35. C	36. A	37. B	38. A	39. B	40. D
41. A	42. C	43. D	44. A	45. C	46. D	47. C	48. A
49. A	50. B	51. D	52. B	53. B	54. C	55. D	56. A
57. C	58. C	59. B	60. D				

Quantitative Skills

61. A	62. D	63. B	64. C	65. A	66. D	67. B	68. B
69. C	70. D	71. A	72. C	73. B	74. C	75. A	76. D
77. B	78. C	79. A	80. B	81. A	82. B	83. D	84. C
85. C	86. D	87. B	88. A	89. D	90. C	91. A	92. B
93. D	94. A	95. C	96. D	97. B	98. C	99. D	100. A
101. D	102. B	103. C	104. D	105. A	106. B	107. C	108. A
109. B	110. D	111. A	112. C				

Reading Comprehension

113. D	114. C	115. C	116. B	117. D	118. C	119. D	120. A
121. B	122. B	123. A	124. A	125. A	126. D	127. C	128. B
129. A	130. C	131. D	132. A	133. A	134. C	135. B	136. B
137. B	138. C	139. D	140. C	141. A	142. D	143. B	144. D
145. C	146. B	147. D	148. C	149. D	150. B	151. D	152. A
153. B	154. C	155. B	156. A	157. D	158. A	159. A	160. C
161. C	162. B	163. B	164. A	165. B	166. C	167. C	168. A
169. D	170. A	171. D	172. B	173. D	174. D		

Mathematics

175. A	176. B	177. B	178. D	179. C	180. C	181. D	182. A
183. C	184. B	185. D	186. D	187. B	188. C	189. A	190. A
191. C	192. B	193. A	194. D	195. B	196. D	197. A	198. A
199. D	200. B	201. C	202. C	203. D	204. B	205. C	206. A

207. D	208. B	209. D	210. A	211. C	212. B	213. A	214. C
215. D	216. B	217. A	218. C	219. B	220. C	221. D	222. A
223. D	224. A	225. D	226. B	227. C	228. B	229. D	230. D
231. C	232. A	233. C	234. C	235. B	236. B	237. A	238. A

Language

239. A	240. B	241. C	242. B	243. D	244. B	245. C	246. C
247. D	248. D	249. A	250. A	251. D	252. A	253. C	254. D
255. B	256. B	257. A	258. D	259. B	260. D	261. B	262. C
263. A	264. C	265. C	266. D	267. A	268. A	269. A	270. D
271. C	272. D	273. B	274. B	275. A	276. C	277. A	278. C
279. C	280. A	281. D	282. B	283. B	284. C	285. C	286. D
287. B	288. D	289. B	290. A	291. D	292. B	293. C	294. A
295. D	296. A	297. C	298. A				

Detailed Explanations

Verbal Skills

1. HSPT-V15-001 - Answer C: walk with heavy, weary steps
To trudge is to walk laboriously with heavy steps, especially when tired, as the hikers are after their climb.

2. HSPT-V15-002 - Answer A: a state of difficulty deciding what to do
A quandary is a difficult state of uncertainty about what to do, fitting the problem of choosing between troublesome alternatives.

3. HSPT-V15-003 - Answer D: reasoning that sounds convincing but is misleading or unsound
Sophistry uses apparently clever reasoning that misleads or conceals a flaw, rather than supplying a genuinely sound argument.

4. HSPT-V15-004 - Answer D: toward the river's mouth
Upstream points toward the source; its opposite is downstream, toward the mouth where the river empties.

5. HSPT-V15-005 - Answer C: having flat, uncreased corners
Dog-eared pages have corners bent or folded over; flat, uncreased corners reverse that particular condition rather than change the pages' contents.

6. HSPT-V15-006 - Answer D: pessimistic
Sanguine describes hopeful confidence about an outcome; pessimistic expresses the opposite expectation that the result is likely to be unfavorable.

7. HSPT-V15-007 - Answer B: break before excessive force damages more valuable parts
Both components protect something more valuable by yielding first: the insert deforms, while the breakaway link separates under excessive force.

8. HSPT-V15-008 - Answer B: time
Plural endings mark grammatical number, while past-tense endings mark when an action occurred rather than ownership, direction, or size.

9. HSPT-V15-009 - Answer B: initial sounds between words
The example exchanges the initial k and t sounds while retaining the remaining sounds, paralleling an exchange of written letters.

10. HSPT-V15-010 - Answer C: sunrise
Sunrise results from a natural cycle; appointments, schedules, and announced deadlines set times through human arrangements.

11. HSPT-V15-011 - Answer A: I; here; today
I identifies the speaker, here the speaker's location, and today the speaking day; every other set includes a term without that direct dependence.

12. HSPT-V15-012 - Answer A: positive; negative, for a real number
Zero is neither positive nor negative, so both descriptions can fail. Each other pair exhausts its stated cases, making exactly one description true.

13. HSPT-V15-013 - Answer B: The speaker treats an unsupported chain of increasingly extreme consequences as inevitable.
The argument jumps from one exception to total loss of rules without establishing the links that would make that escalation inevitable.

14. HSPT-V15-014 - Answer A: The total number of entries equals the total number of exits.
Each entry changes outside to inside and needs a later exit to finish outside; pairing these changes balances total entries and exits across both gates.

15. HSPT-V15-015 - Answer C: Permission alone does not establish which sessions the holder actually attended.
The pass specifies available permissions, not completed actions; without a choice or attendance record, actual attendance remains unknown.

16. HSPT-V15-016 - Answer D: worn and exhausted
Haggard describes a tired, worn appearance, often resulting from worry or lack of sleep, as in the sailor's situation.

17. HSPT-V15-017 - Answer D: capable of being pressed into a new shape
Malleable material can be hammered or pressed into shape without breaking; the context concerns shaping rather than heat or magnetism.

18. HSPT-V15-018 - Answer A: persuade with coaxing or flattery
To cajole is to coax someone into acting, often through gentle persuasion or flattery rather than threats or physical force.

19. HSPT-V15-019 - Answer B: courteous
Brusque behavior is abrupt and blunt; courteous behavior shows consideration and politeness toward another person.

20. HSPT-V15-020 - Answer D: enthusiastic
Tepid support lacks energy or enthusiasm; enthusiastic support expresses the strong commitment that the original description lacks.

21. HSPT-V15-021 - Answer A: steadfast
Fickle describes readily changing loyalties or preferences; steadfast describes remaining firm and loyal despite opportunities to change.

22. HSPT-V15-022 - Answer C: constraining material within a chosen shape
Both tools produce a desired form through physical limits: the stencil restricts coverage, while the mold restricts the material's shape.

23. HSPT-V15-023 - Answer B: A space separating neighboring words
A musical rest makes a meaningful interval without sound; a word space makes a meaningful interval without printed letters.

24. HSPT-V15-024 - Answer D: Drawing a recognizable outline while leaving out an object's fine details
Both retain enough identifying information for recognition while deliberately omitting detail; replacement, rearrangement, and exact copying do not.

25. HSPT-V15-025 - Answer B: bolt
A bolt is a piece of fastening hardware; a bevel, groove, and scratch describe surface shapes or marks.

26. HSPT-V15-026 - Answer B: applaud
Authorize, permit, and license can grant permission; applaud expresses praise or approval without necessarily granting permission.

27. HSPT-V15-027 - Answer C: junction
A junction is a meeting place of distinct routes or parts; the other choices name inner portions rather than interfaces.

28. HSPT-V15-028 - Answer C: Its address may have been checked, but the missing tab does not establish that.
A checked package can lose its tab, while an unchecked package lacks one; the present absence therefore does not settle its history.

29. HSPT-V15-029 - Answer C: Orchard is immediately after Harbor.
The consecutive Drift–Harbor block must precede Orchard, which must precede Summit, forcing the complete order Drift, Harbor, Orchard, Summit.

30. HSPT-V15-030 - Answer A: The latch is not brass, but the door's color remains undetermined.
Both statements are false; a false brass claim already makes the conjunction false, whatever the door's color may be.

31. HSPT-V15-031 - Answer B: pale
Pallid describes a pale appearance lacking normal color; it does not by itself describe an emotion or a sunburn.

32. HSPT-V15-032 - Answer C: a point at which progress is blocked
An impasse is a deadlock where progress cannot continue, fitting talks in which neither side will alter its position.

33. HSPT-V15-033 - Answer D: grow weak or fail to prosper
Languish means become weak or fail to thrive; the lack of maintenance and customers supports that sense in the sentence.

34. HSPT-V15-034 - Answer D: fine
Coarse fabric has a rough texture or thick fibers; fine fabric has a delicate, smoother texture, reversing the intended meaning.

35. HSPT-V15-035 - Answer C: definite
A tentative date is provisional and open to revision; a definite date is firmly settled rather than merely proposed.

36. HSPT-V15-036 - Answer A: devoid of
Replete with means filled or abundantly supplied with something; devoid of means entirely lacking it.

37. HSPT-V15-037 - Answer B: A pronunciation key explains the sound symbols in a dictionary.
Both a map legend and a pronunciation key explain how a reader should interpret symbols used in a larger reference work.

38. HSPT-V15-038 - Answer A: reveals design problems before a product is manufactured widely
Rehearsals and prototypes permit problems to be discovered during a trial stage before the final activity is presented or produced widely.

39. HSPT-V15-039 - Answer B: Repairing a damaged device; discarding the device entirely
Correction and repair preserve the original with a fault addressed; retraction and discarding remove the original from continued use.

40. HSPT-V15-040 - Answer D: A parcel's tracking code
A tracking code identifies a particular parcel by assignment; weight, height, and temperature quantify physical properties.

41. HSPT-V15-041 - Answer A: dialogue; discussion; negotiation
Dialogue, discussion, and negotiation involve participants exchanging information; broadcasts, announcements, and monologues can communicate in only one direction.

42. HSPT-V15-042 - Answer C: They are reserves kept for a possible future need.
Each item is deliberately held available for a possible later need, rather than being leftover waste or a reward.

43. HSPT-V15-043 - Answer D: The design has not met either approval route.
The curator has not approved, and the joint route lacks the carpenter's approval; therefore neither stated route has been satisfied.

44. HSPT-V15-044 - Answer A: Exactly one of the two gates is open.
Both open would make all statements true, and both closed would make all false; either single open gate makes exactly one statement true.

45. HSPT-V15-045 - Answer C: A timer independently starts the fan and switches off the lamp at the same moment.
A common timer could produce both observed events, so their repeated association would not establish that the fan causes the lamp's change.

46. HSPT-V15-046 - Answer D: search for food
To forage is to search for food, which fits birds exploring exposed mud for something to eat.

47. HSPT-V15-047 - Answer C: firmly loyal
Staunch describes firm, dependable loyalty; Imani continues supporting the proposal even after other supporters leave.

48. HSPT-V15-048 - Answer A: boldly daring
Audacious means boldly daring or willing to take risks, matching a surprising proposal presented to a cautious panel.

49. HSPT-V15-049 - Answer A: incautious
Wary describes caution and alertness to risk; incautious describes a lack of that caution rather than a different kind of vigilance.

50. HSPT-V15-050 - Answer B: undemanding
Onerous work imposes a heavy burden; undemanding work requires little effort and therefore reverses the intended meaning.

51. HSPT-V15-051 - Answer D: consequential
Trivial means unimportant in this context; consequential describes something important because of its effects or implications.

52. HSPT-V15-052 - Answer B: Choreography specifies how dance movements are arranged.
Both provide an organized plan for arranging individual elements into a larger work: stitches in knitting and movements in dance.

53. HSPT-V15-053 - Answer B: Adjusting its throttle to change its output
A dimmer and a throttle regulate the level of an ongoing operation; the alternatives stop operation or change something unrelated to output.

54. HSPT-V15-054 - Answer C: A narrow passage limits traffic arriving from several wide roads.
Both systems have a bottleneck: increasing capacity elsewhere does not remove the limit imposed by their narrowest or slowest stage.

55. HSPT-V15-055 - Answer D: marks left by physical contact
Each is a physical mark produced by contact or pressure, rather than a measurement, weather term, or sound.

56. HSPT-V15-056 - Answer A: vacant; occupied; reserved
Vacant, occupied, and reserved describe availability; polished, heavy, bright, and wide describe other physical characteristics.

57. HSPT-V15-057 - Answer C: A verbatim transcript
A verbatim transcript records the words as spoken; an abstract, synopsis, or overview selects and condenses information.

58. HSPT-V15-058 - Answer C: The fuel check occurs before launch, but its order relative to the radio check is not fixed.
Fuel precedes crew, which precedes launch; radio also precedes launch, but no rule places it before or after the fuel check.

59. HSPT-V15-059 - Answer B: At least one description in that log entry is incorrect.
A truck cannot qualify as electric, so northbound travel requires a permit; the entry cannot satisfy both rules as written.

60. HSPT-V15-060 - Answer D: Disproving both rules would require at least two tiles, one from each group.
A counterexample to Rule 1 must be glazed, while a counterexample to Rule 2 must be unglazed; a single tile cannot be both.

Quantitative Skills

61. HSPT-Q15-001 - Answer A: 4
Starting at 16, the integer halvings are 16 → 8 → 4 → 2 → 1. Four halvings are possible before reaching an odd integer.

62. HSPT-Q15-002 - Answer D: 4
One item remains after minute four. Five more arrive, making six waiting; processing two leaves four items.

63. HSPT-Q15-003 - Answer B: 110
In 111, the original first and last digits match, so append 0 after shifting the last two digits left: 110.

64. HSPT-Q15-004 - Answer C: 6
For four beads: two single-color patterns, two patterns with one bead of the minority color, and two with two of each color. Total: six.

65. HSPT-Q15-005 - Answer A: −7
The larger negative number is −2, and the smaller positive number is 5. Subtracting gives −2 − 5 = −7.

66. HSPT-Q15-006 - Answer D: 56
Split the string as 101 and 110. The first group gives 4 + 1 = 5; the second gives 4 + 2 = 6, producing 56.

67. HSPT-Q15-007 - Answer B: 4
The pairs are (7, 42), (8, 24), (9, 18), and (10, 15). Since a < b, 6 < a < 12; checking those five integers gives four pairs.

68. HSPT-Q15-008 - Answer B: Quantity B is greater.
Since a + b + c = 3m, Quantity A expands to $a^2 + b^2 + c^2 - 3m^2$. The subtracted amount is strictly positive.

69. HSPT-Q15-009 - Answer C: The two quantities are equal.
Absolute value removes any product sign. Taking each factor's absolute value before multiplication produces the same nonnegative magnitude, including when a factor is zero.

70. HSPT-Q15-010 - Answer D: The relationship cannot be determined from the information given.
For 1, 1, 4, the means are 2.5 and 2; for 1, 4, 4, they are 2.5 and 3. Removing duplicates can change the ordering either way.

71. HSPT-Q15-011 - Answer A: Quantity A is greater.
The interior point makes one triangle along each of n sides. Diagonals from one vertex divide the polygon into n − 2 triangles, so Quantity A is greater.

72. HSPT-Q15-012 - Answer C: The two quantities are equal.
The original angles sum to 180°. Their adjacent halves sum to 180° ÷ 2 = 90°, giving perpendicular bisecting rays.

73. HSPT-Q15-013 - Answer B: Quantity B is greater.
For common perimeter P, the square's area is $P^2/16$ and the circle's is $P^2/(4\pi)$. Since π < 4, the circle has the larger area.

74. HSPT-Q15-014 - Answer C: 27
Reverse 63 to obtain 36; subtracting the smaller from the larger gives 63 − 36 = 27.

75. HSPT-Q15-015 - Answer A: 1
The last pair is 4 and 3; dividing 4 by 3 leaves remainder 1, which is the next entry.

76. HSPT-Q15-016 - Answer D: 41/29
For 17/12, the numerator becomes 41/12 and the denominator becomes 29/12; their quotient is 41/29.

77. HSPT-Q15-017 - Answer B: 4
Seventeen positions contain five complete three-digit blocks and two more digits; the second digit of block 246 is 4.

78. HSPT-Q15-018 - Answer C: 99
The multiples of 9 in the interval are 72, 81, 90, and 99; only 99 is one less than a multiple of 5.

79. HSPT-Q15-019 - Answer A: 3.50
Rounding to 4 requires at least 3.50; rounding to 3.5 permits values through 3.54, so the smallest qualifying hundredth is 3.50.

80. HSPT-Q15-020 - Answer B: 7
For a = 1 there are four triples; for a = 2 there are two; for a = 3 there is one, totaling seven.

81. HSPT-Q15-021 - Answer A: Quantity A is greater.
Multiplying both quantities by positive 1 − x gives 1 + x and $1 + x - 2x^2$; the first is greater.

82. HSPT-Q15-022 - Answer B: Quantity B is greater.
Writing k = 4m + 1 gives $k^2 = 16m^2 + 8m + 1$, so Quantity A is 1, less than 3.

83. HSPT-Q15-023 - Answer D: The relationship cannot be determined from the information given.
The numbers −2, 1, 1 give product −2, while 2, −1, −1 give product 2; both sets satisfy the required zero sum.

84. HSPT-Q15-024 - Answer C: The two quantities are equal.
The triangles have equal bases BD and DC and share the same perpendicular height from A to line BC.

85. HSPT-Q15-025 - Answer C: The two quantities are equal.
A cone has one third the volume of a cylinder with the same base area and height, making the stated quantities equal.

86. HSPT-Q15-026 - Answer D: The relationship cannot be determined from the information given.
A right included angle gives area 17.5, while a 30-degree included angle gives 8.75; equality and a smaller area are both possible.

87. HSPT-Q15-027 - Answer B: 5
The given counts correspond to one through six; the next name, seven, has five letters.

88. HSPT-Q15-028 - Answer A: 59
The digits of 52 sum to 7, so the next entry is 52 + 7 = 59.

89. HSPT-Q15-029 - Answer D: 44/9
The preceding three terms sum to 6 + 4 + 14/3 = 44/3; dividing by 3 gives 44/9.

90. HSPT-Q15-030 - Answer C: 6
For 6 units, use six 1s, two 3s, or one 3 with three 1s in four orders: 1 + 1 + 4 = 6.

91. HSPT-Q15-031 - Answer A: 603
An odd number must end in 3; placing 6 in the hundreds position and 0 in the tens position gives 603.

92. HSPT-Q15-032 - Answer B: 20
Since 45 = $3^2 \times 5$, k needs a factor 5 and an even exponent of 2; the smallest even choice is $2^2 \times 5 = 20$.

93. HSPT-Q15-033 - Answer D: 8
Every sum is 11 times a digit sum from 3 through 17; the palindromes are 33, 44, 55, 66, 77, 88, 99, and 121.

94. HSPT-Q15-034 - Answer A: Quantity A is greater.
Subtracting Quantity B from Quantity A gives $ab - a - b + 1 = (a - 1)(b - 1)$, which is positive.

95. HSPT-Q15-035 - Answer C: The two quantities are equal.
The two middle observations are the smallest positive value and its negative; their average, the median, is 0.

96. HSPT-Q15-036 - Answer D: The relationship cannot be determined from the information given.
If the events always occur together, their joint probability is 1/2; if they are mutually exclusive, it is 0, so either comparison is possible.

97. HSPT-Q15-037 - Answer B: Quantity B is greater.
The diagonal is shorter than the two-side route between opposite corners, so $d < 2s$ and therefore $d - s < s$.

98. HSPT-Q15-038 - Answer C: The two quantities are equal.
Four cube faces have combined area $4s^2$; the larger square has area $(2s)^2 = 4s^2$.

99. HSPT-Q15-039 - Answer D: The relationship cannot be determined from the information given.
An apex above the base midpoint gives perimeter $4 + 4\sqrt{2} < 10$; above a base endpoint it gives $6 + 2\sqrt{5} > 10$.

100. HSPT-Q15-040 - Answer A: 3
The next move is 5 positions from 4; passing 5 wraps to 0, so the destination is 3.

101. HSPT-Q15-041 - Answer D: 32
Multiplying 21 by 1.5 gives 31.5; the stated halfway rule rounds that value up to 32.

102. HSPT-Q15-042 - Answer B: 3
After 5, the next remainder is 1, so the six-entry cycle repeats; entry 20 matches entry 2 because $20 = 3 \times 6 + 2$.

103. HSPT-Q15-043 - Answer C: 11
The first ten labels are 1, 2, 3, 5, 6, 7, 8, 9, 10, and 11.

104. HSPT-Q15-044 - Answer D: 2
For input x, Machine A gives $(2x)^2 = 4x^2$, while Machine B gives $2x^2$; their ratio is 2.

105. HSPT-Q15-045 - Answer A: 2
Subtracting x from 100x gives $99x = 18$, so $x = 2/11$ and therefore $11x = 2$ exactly.

106. HSPT-Q15-046 - Answer B: 36
The possible unordered pairs give products 12, 22, 30, 36, 40, and 42; only 12 and 36 are divisible by 12.

107. HSPT-Q15-047 - Answer C: The two quantities are equal.
Expanding and subtracting gives $x^2 + 2x + 1 - (x^2 - 2x + 1) = 4x$ for every real x.

108. HSPT-Q15-048 - Answer A: Quantity A is greater.
The quantities equal (b − a)/a and (b − a)/b; their common positive numerator divided by the smaller denominator gives the larger value.

109. HSPT-Q15-049 - Answer B: Quantity B is greater.
Every entry is at most n and at least one is smaller, so their arithmetic mean is strictly less than n.

110. HSPT-Q15-050 - Answer D: The relationship cannot be determined from the information given.
Boxes of dimensions 2 by 2 by 2 and 1 by 1 by 8 both have volume 8, but edge totals 24 and 40; either box can be first.

111. HSPT-Q15-051 - Answer A: Quantity A is greater.
For edge s, the interior diagonal is $s\sqrt{3}$ and a face diagonal is $s\sqrt{2}$; since $\sqrt{3} > \sqrt{2}$, Quantity A is greater.

112. HSPT-Q15-052 - Answer C: The two quantities are equal.
Joining the center to consecutive vertices forms six equilateral triangles, so each hexagon side equals r and its perimeter equals 6r.

Reading Comprehension

113. HSPT-R15-001 - Answer D: The tip of the post's shadow
The opening places the nut at the tip of the long shadow, which the bird treats as a finger pointing to its cache.

114. HSPT-R15-002 - Answer C: Disturbed soil beside a band of yellow leaves
The bird notices previously disturbed soil near the yellow leaves and scratches there, uncovering the nut in its original place.

115. HSPT-R15-003 - Answer C: It treats the tip of a changing shadow as though it marks one fixed location.
The nut remains in its original hole, while the shorter shadow ends elsewhere; relying on that moving tip directs the bird to new ground.

116. HSPT-R15-004 - Answer B: marked with cuts or grooves
The deep notches are physical marks in the root, so scored means cut or grooved rather than evaluated, counted, or notated musically.

117. HSPT-R15-005 - Answer D: It expresses the jay's confidence in a pointer whose changing position readers can recognize as unreliable.
The finger image makes the shadow seem like a helpful guide from the bird's perspective, setting up the contrast when its tip shifts.

118. HSPT-R15-006 - Answer C: The weight of the water displaced by the vessel
The first paragraph directly defines buoyant force as equal to the weight of the water the vessel displaces.

119. HSPT-R15-007 - Answer D: Its overall shape can displace water of sufficient weight to balance the vessel before the hull is fully submerged.
The passage distinguishes steel's material density from the hollow vessel's overall displacement, which can provide enough upward force to balance its weight.

120. HSPT-R15-008 - Answer A: They displace water of equal weight.
Each floating vessel must displace water whose weight balances its own; equal vessel weights therefore require equal displaced-water weights.

121. HSPT-R15-009 - Answer B: comes to rest at a lower level
The added load makes the vessel take a lower balanced position in the water, so settles concerns its resulting physical level.

122. HSPT-R15-010 - Answer B: Keeping the material the same helps isolate the effect of overall shape and displaced volume.
Both examples use steel, so their contrasting behavior highlights shape and displacement rather than a change from one material to another.

123. HSPT-R15-011 - Answer A: Each has one identical slip, folded alike and mixed with the others.
The opening specifies one identical slip for each household, equal folding, and mixing before a single slip is drawn.

124. HSPT-R15-012 - Answer A: A draw can resolve a tie among appropriately equal claims, but it cannot justify treating the claims as equal in the first place.
The author separates fair selection within an established tie from the reasoned judgment needed to decide whether the claims deserve equal standing.

125. HSPT-R15-013 - Answer A: Whether the difference in need justifies assigning different priority.
The unequal-needs example requires a reasoned decision about priority before the households can be treated as equally eligible for the draw.

126. HSPT-R15-014 - Answer D: justified
The sentence requires reasons supporting equal priority, so warranted means justified rather than financed, postponed, or randomly selected.

127. HSPT-R15-015 - Answer C: The difference between selecting an outcome and justifying the claims that may enter the selection.
The parallel can and cannot clauses sharpen the boundary between what random selection accomplishes and what still requires reasons.

128. HSPT-R15-016 - Answer B: It qualifies the initial case by showing that equal representation is appropriate only after priority has been considered.
The new case limits an overgeneralization from the original tie: a fair draw does not replace judgment about whether claims should be ranked equally.

129. HSPT-R15-017 - Answer A: A red cushion
The boy places his red cushion on the bare floor beneath the window where his reading chair may eventually stand.

130. HSPT-R15-018 - Answer C: Seeing another family begin to imagine the room helps the narrator accept leaving it.
The narrator moves from wanting the room unchanged to accepting its different future after watching the boy establish a place for himself.

131. HSPT-R15-019 - Answer D: An ordinary physical mark carries a personal memory that makes departure difficult.
The birthday memory connects a minor wall mark to the narrator's earlier life, showing why the room feels personally significant.

132. HSPT-R15-020 - Answer A: It personifies the walls to describe the echo of the narrator's voice.
Walls cannot literally hand back a name; the human action makes the returning sound of the echo vivid.

133. HSPT-R15-021 - Answer A: It lets readers compare the narrator's private resistance with the narrator's later willingness to leave.
Readers know the narrator's earlier wish and final change directly, while the other characters' private thoughts remain unavailable.

134. HSPT-R15-022 - Answer C: Many tiny spaces created during expansion
The passage says vapor creates many tiny spaces in softened starch and those spaces remain when the expanded starch firms.

135. HSPT-R15-023 - Answer B: Heat, internal water, a pressure-retaining covering, and softened starch cooperate to produce rapid expansion.
The passage links the opening to interacting conditions and explains each condition's contribution rather than attributing the result to heat alone.

136. HSPT-R15-024 - Answer B: The crack may let vapor escape before sufficient pressure accumulates.
The passage explains that a damaged covering can release vapor early, so identical heating does not ensure equal pressure buildup.

137. HSPT-R15-025 - Answer B: keeps within limits
The covering confines vapor by keeping it inside temporarily, allowing pressure to rise until the covering breaks.

138. HSPT-R15-026 - Answer C: Their failures show why the conditions in the explanation are necessary.
The unsuccessful cases remove either effective containment or sufficient water, testing the roles assigned to those conditions in the preceding explanation.

139. HSPT-R15-027 - Answer D: At the left side of the wheel
The example explicitly places the mark at the left after a three-quarter clockwise turn from its initial top position.

140. HSPT-R15-028 - Answer C: Enlarging the existing pictures
The final paragraph states that enlarging existing pictures cannot add the intervening positions that were never recorded.

141. HSPT-R15-029 - Answer A: Separate recorded positions can make continuous motion appear to have the wrong direction.
The mark example and sampling discussion explain how missing intermediate positions can lead viewers to infer an incorrect direction.

142. HSPT-R15-030 - Answer D: The wheel could be still or could complete whole revolutions between pictures.
The passage notes that complete revolutions between samples go unrecorded, so repeated positions do not distinguish rest from those rotations.

143. HSPT-R15-031 - Answer B: separate observations taken at particular times
The samples are the individual recorded pictures, each observing the wheel at a particular time rather than continuously.

144. HSPT-R15-032 - Answer D: It contrasts a change that can reduce the information gap with a change that cannot.
More frequent pictures can reduce missing motion, whereas enlargement cannot recover it; the contrast clarifies the source of the difficulty.

145. HSPT-R15-033 - Answer C: Platform Four
After Suri asks for help, the attendant explicitly directs her to Platform Four before she boards a tram.

146. HSPT-R15-034 - Answer B: Seeking needed information is more useful than trying to appear already knowledgeable.
Suri first protects an appearance of certainty, then finds her way by admitting uncertainty and asking a specific question.

147. HSPT-R15-035 - Answer D: She is looking for a reason to keep her first choice without admitting that she needs help.
Her unsupported guess follows an unverified claim of certainty; checking the listed stops reveals that the guess does not solve her problem.

148. HSPT-R15-036 - Answer C: trying to look like an experienced traveler without yet feeling like one
The coat gives Suri an appearance associated with her confident cousin, while her uncertainty exposes the gap between that appearance and her knowledge.

149. HSPT-R15-037 - Answer D: The playful personification highlights her decision to rely on useful actions rather than an appearance.
The coat cannot literally know a route; the humorous wording exposes Suri's earlier reliance on looking knowledgeable instead of obtaining information.

150. HSPT-R15-038 - Answer B: The narrator is supposed to open the jar marked THUNDEROUS.
The narrator explains that two raised fingers are the signal for opening the larger jar labeled THUNDEROUS.

151. HSPT-R15-039 - Answer D: Organized sounds of approval differ from freely felt and expressed appreciation.
The controlled bottled claps contrast with the audience's spontaneous response, which finally makes the stored applause unnecessary.

152. HSPT-R15-040 - Answer A: Opening it would replace or cover the genuine response already developing in the room.
The narrator hears spontaneous applause and chooses to listen instead of adding the mayor's manufactured response over it.

153. HSPT-R15-041 - Answer B: His system directs attention toward his signals instead of encouraging people to respond to what deserves admiration.
Although the mayor claims to organize proper appreciation, his audience watches his hand rather than attending freely to the music.

154. HSPT-R15-042 - Answer C: It establishes an expected routine that the narrator later chooses to interrupt.
The opening makes the signal-and-jar routine clear, allowing the narrator's final refusal to mark a meaningful departure from it.

155. HSPT-R15-043 - Answer B: The slider
The passage identifies the slider's converging internal channel as the component that guides and aligns the existing teeth.

156. HSPT-R15-044 - Answer A: Matching teeth form many small connections as the slider guides them together or apart.
Both opening and closing depend on guided engagement of existing matching teeth, rather than new stitches, manufactured teeth, or adhesive.

157. HSPT-R15-045 - Answer D: Guiding the rows cannot supply the connecting structure that is absent.
The passage distinguishes guiding intact teeth from replacing them; a missing tooth remains absent even if the guiding action works normally.

158. HSPT-R15-046 - Answer A: a fitted connection that holds the parts together
Here engagement describes the matching teeth fitting and holding together inside the closure, rather than a social promise or an occupation.

159. HSPT-R15-047 - Answer A: Opening the zipper by guiding its rows apart
The next paragraph begins by stating that opening reverses the guided route, then describes the two rows separating.

160. HSPT-R15-048 - Answer C: Wanting to draw the crane
The narrator opens the sketchbook and explicitly calls out a wish to draw the crane when Uncle Pavo arrives.

161. HSPT-R15-049 - Answer C: He accepts a reason for making the stop last longer without discussing his need to rest.
He sees only a few quick lines, then emphasizes a time-consuming detail and invites the narrator to take time, quietly accepting the extended pause.

162. HSPT-R15-050 - Answer B: done with noticeable effort
The widening gap and the uncle's later pause show that labored describes pedaling requiring increased effort.

163. HSPT-R15-051 - Answer B: It creates a parallel in which the person quietly making room for the other's slower pace has changed.
The uncle once supplied an excuse for waiting; the narrator now supplies one, making their exchanged roles visible through the parallel.

164. HSPT-R15-052 - Answer A: Their consideration for each other emerges through actions and hints instead of an explicit announcement.
The pause, the rope remark, and the recalled excuses communicate care indirectly, preserving the quiet understanding between the characters.

165. HSPT-R15-053 - Answer B: A local change in an electric field
The first paragraph says a conductive finger changes the nearby electric field and that the electronics detect that change.

166. HSPT-R15-054 - Answer C: A glove must allow the kind of interaction a screen detects, so contact or flexibility alone may not be enough.
The passage connects glove performance to electrical properties on capacitive screens while distinguishing other sensing mechanisms.

167. HSPT-R15-055 - Answer C: Its electrical properties provide an interaction that the other glove does not provide.
With thickness and flexibility held alike, the passage supports electrical interaction as the relevant difference rather than pressure or a changed screen mechanism.

168. HSPT-R15-056 - Answer A: appropriate for the required purpose
Suitable describes a connection appropriate for enabling the electrical interaction the screen needs, rather than an appearance or color.

169. HSPT-R15-057 - Answer D: To limit the explanation's scope and clarify why results can differ between screen types.
The final paragraph distinguishes capacitive screens from other designs, preventing the preceding account from being applied indiscriminately to every touchscreen.

170. HSPT-R15-058 - Answer A: To track payments
The first paragraph expressly distinguishes the book's purpose of tracking payments from describing all activities at the inn.

171. HSPT-R15-059 - Answer D: A historical record should be interpreted in light of what it was created to document.
The passage repeatedly connects what the ledger includes or omits to its payment-tracking purpose and urges interpretation within that limit.

172. HSPT-R15-060 - Answer B: A letter states that a neighbor stopped giving the inn vegetables that year.
Losing a source of gifts could require more purchases without any increase in vegetables consumed, directly supporting the passage's alternative explanation.

173. HSPT-R15-061 - Answer D: piece together from available evidence
The later reader uses surviving records to piece together a broader account of past events, rather than inventing or merely copying it.

174. HSPT-R15-062 - Answer D: It preserves the ledger's value while defining the kinds of conclusions it can support most directly.
The conclusion says incompleteness does not make the book useless and redirects attention to the transactions it was designed to record.

Mathematics

175. HSPT-M15-001 - Answer A: 1
Putting n = 1 into the rule gives 1! = 1 × 0!. Since 1! is 1, 0! must also be 1.

176. HSPT-M15-002 - Answer B: It is even.
The new fraction is (2a + b)/(2b). Its numerator is odd, so reducing cannot cancel the denominator's factor of 2.

177. HSPT-M15-003 - Answer B: 60
Exponent-count patterns 12, 6 × 2, 4 × 3, and 3 × 2 × 2 have respective minimum integers 2048, 96, 72, and 60. The least is 60.

178. HSPT-M15-004 - Answer D: 3
First find |3 − 8| = 5. Then take |2 − 5| = |−3| = 3; both absolute-value operations must be applied.

179. HSPT-M15-005 - Answer C: x/a
With a fixed, x/a equals (1/a)x, a constant multiple of x. The other expressions contain x squared or in a denominator.

180. HSPT-M15-006 - Answer C: a < 0
The inequality is 2a + b > 5a + b. Cancel b to obtain −3a > 0, which requires a < 0.

181. HSPT-M15-007 - Answer D: 4
The two distinct integer outputs f(2) and f(3) must lie strictly between 2 and 5. They must therefore be 3 and 4, respectively.

182. HSPT-M15-008 - Answer A: p = s/10 + 5
Because t = s/60, substitute to get p = 6(s/60) + 5 = s/10 + 5. The position unit and initial position remain unchanged.

183. HSPT-M15-009 - Answer C: 7 centimeters
A parallelogram's diagonals bisect each other. Thus O is the midpoint of AC, and OC equals AO, or 7 centimeters.

184. HSPT-M15-010 - Answer B: 36 square units
The parallel cross-section is a square with half the base's side length, 6 units. Its area is $6^2 = 36$ square units.

185. HSPT-M15-011 - Answer D: 54 square units
The overlap is a regular hexagon with side one-third of a triangle's side. Its six small triangles each have 1/9 of the original area: 6 × 81/9 = 54.

186. HSPT-M15-012 - Answer D: 45 seconds
The excerpt contains 24 × 3 = 72 beats. At 96 beats per minute, it lasts 72/96 minute, or 45 seconds.

187. HSPT-M15-013 - Answer B: 9 < A < 15
The full squares contribute 9. The six partial squares add more than 0 but less than 6, giving an area strictly between 9 and 15.

188. HSPT-M15-014 - Answer C: Random assignment to treatment groups
The coin allocates existing volunteers to methods; it does not select people from the wider population or measure their performance.

189. HSPT-M15-015 - Answer A: 18/67
There are 20 cracked parts, of which 18 are flagged. Of 980 uncracked parts, 49 are flagged. The required probability is 18/(18 + 49) = 18/67.

190. HSPT-M15-016 - Answer A: 3/8
Choose two of the three boundaries for changes, in three ways, and choose the first color, in two ways. Six of 16 equally likely patterns work.

191. HSPT-M15-017 - Answer C: 16
Thirty-six percent of 25 is 9 colored squares; subtracting 9 from 25 leaves 16 uncolored squares.

192. HSPT-M15-018 - Answer B: 1/5
The materials contribute 3 × 1/6 + 2 × 1/4 = 1 kilogram of copper in 5 kilograms of alloy, giving 1/5.

193. HSPT-M15-019 - Answer A: 1/6 + 1/3
The sum 1/6 + 1/3 reduces to 1/2 = 0.5; the other sums reduce to 4/9, 5/18, and 5/12, which repeat.

194. HSPT-M15-020 - Answer D: A fixed charge independent of the number of hours
The term 9 does not change with h, so it represents a fixed charge added to the hourly cost.

195. HSPT-M15-021 - Answer B: 4x + 2 centimeters
There are x − 1 joints, so the finished length is 6x − 2(x − 1) = 4x + 2 centimeters.

196. HSPT-M15-022 - Answer D: 1
Expanding both squares and canceling x^2 gives −8x + 16 = 4x + 4, so 12x = 12 and x = 1.

197. HSPT-M15-023 - Answer A: 2, 7, and 24
A final 6 can follow intermediate 12 or 1; their positive predecessors are respectively 24 or 7, and 2.

198. HSPT-M15-024 - Answer A: (3, −5)
Reflection across the x-axis changes 5 to −5; reflection across the y-axis then changes −3 to 3.

199. HSPT-M15-025 - Answer D: 12 centimeters
A trapezoid's midpoint segment has length equal to the average of its bases: (8 + 16)/2 = 12 centimeters.

200. HSPT-M15-026 - Answer B: 20 + 8√2 units
The original perimeter is 36; the cuts remove 16 units of edge and add four diagonals of length 2√2, giving 20 + 8√2.

201. HSPT-M15-027 - Answer C: 35 milliliters
Two tablespoons equal 6 teaspoons; adding one gives 7 teaspoons, and 7 × 5 = 35 milliliters.

202. HSPT-M15-028 - Answer C: 18
The remaining mass is 1,800 × 0.82 = 1,476 grams; dividing by 82 grams per portion gives exactly 18 portions.

203. HSPT-M15-029 - Answer D: 07:15 Tuesday
The destination is 5 hours ahead: departure is 03:40 Tuesday there, and adding 3 hours 35 minutes gives 07:15 Tuesday.

204. HSPT-M15-030 - Answer B: 6 centimeters
The range is 20 − 7 = 13 centimeters, and the mode is 7 centimeters; their difference is 6 centimeters.

205. HSPT-M15-031 - Answer C: 2/5
There are 20 three-card sets; a divisible sum requires one card from each remainder group {1,4}, {2,5}, {3,6}, giving 8 favorable sets.

206. HSPT-M15-032 - Answer A: Median
The middle value is 12 before removal and remains 12 in the five-value list; the mean, range, and observation count all change.

207. HSPT-M15-033 - Answer D: 1.5 grams and 2 grams
The fraction 15/8 equals 1.875, which is greater than 1.5 and less than 2.

208. HSPT-M15-034 - Answer B: $1.25
Invoice A is 0.75p + 5; Invoice B is 0.75(p + 5) = 0.75p + 3.75, so A charges $1.25 more.

209. HSPT-M15-035 - Answer D: −19/24
Writing the bounds as −20/24 and −18/24 places −19/24 strictly between them; the other choices lie outside the interval.

210. HSPT-M15-036 - Answer A: 31/42
The interval length is 13/6, so each part is 13/42; moving four parts from −1/2 gives −21/42 + 52/42 = 31/42.

211. HSPT-M15-037 - Answer C: 3
The denominator becomes zero when x = 3, and division by zero is undefined regardless of the nonzero numerator.

212. HSPT-M15-038 - Answer B: 29
The first two terms equal 3(2a + 3b) = 33; subtracting 4 gives 29 without needing either variable separately.

213. HSPT-M15-039 - Answer A: 4
The permitted extra mass is 50 − 3 − 12 = 35 kilograms; 35 ÷ 7.5 is between 4 and 5, allowing four complete parcels.

214. HSPT-M15-040 - Answer C: 4
The inequalities give n ≥ 6 and n < 10.5; being able to remove 7 also requires n ≥ 7, leaving 7, 8, 9, and 10.

215. HSPT-M15-041 - Answer D: The center
The center is the pivot of the rotation, so it remains fixed while every specified point away from it moves.

216. HSPT-M15-042 - Answer B: 35 square centimeters
The triangular base's longest side is √(3² + 4²) = 5; the largest rectangular face therefore has area 5 × 7 = 35.

217. HSPT-M15-043 - Answer A: 16 square units
Base BD is half of BA, and E is one third as high above line AB as C; the area is 96 × 1/2 × 1/3 = 16.

218. HSPT-M15-044 - Answer C: 375 liters
The meter records a change of 3.125 − 2.750 = 0.375 cubic meter; multiplying by 1,000 gives 375 liters.

219. HSPT-M15-045 - Answer B: 20 meters
The original plan length was 12.5 ÷ 1.25 = 10 centimeters; at 1:200 that represents 2,000 centimeters, or 20 meters.

220. HSPT-M15-046 - Answer C: People who choose to respond may differ from those who do not.
Voluntary participation can select people with different experiences or motivation from nonrespondents, even when many responses are collected.

221. HSPT-M15-047 - Answer D: 2/3
The qualifying pairs are (2,1), (3,1), (3,2), (4,1), (4,2), and (4,3); four of these six have an even left number.

222. HSPT-M15-048 - Answer A: Individual rounding can explain the displayed total of 101% without an error in the underlying counts.
Exact percentages of 33.6%, 33.6%, and 32.8% total 100% but round to the displayed figures, so rounding alone can explain the total.

223. HSPT-M15-049 - Answer D: The second rate is lower by 5 percentage points.
The rates are 5/20 = 25% and 8/40 = 20%; their difference is 5 percentage points despite the larger second defect count.

224. HSPT-M15-050 - Answer A: −2°C
The morning increase is 16°C; three fourths of that is 12°C, so the evening temperature is 10 − 12 = −2°C.

225. HSPT-M15-051 - Answer D: 8
Seven pairs of 2 and 5 make seven factors of 10, leaving three factors of 2; the number is 80,000,000, which has eight digits.

226. HSPT-M15-052 - Answer B: −(x + 1)
When x is positive, x + 1 is positive and its negative is always negative; x − 1 does not keep a negative sign for every x.

227. HSPT-M15-053 - Answer C: 3
Multiplying by x − 2 gives x = 3x − 6, so 2x = 6 and x = 3, which satisfies the domain restriction.

228. HSPT-M15-054 - Answer B: 11/2
Composing gives g(g(t)) = 2(2t − 5) − 5 = 4t − 15; setting this equal to 7 yields t = 11/2.

229. HSPT-M15-055 - Answer D: 7
Six clips last at most 30 minutes; seven clips can total 31 using two 3-minute clips and five 5-minute clips, so seven is minimal.

230. HSPT-M15-056 - Answer D: Its area
Reflection preserves lengths and area, though it can change horizontal coordinates, reverse left-to-right order, and move the figure.

231. HSPT-M15-057 - Answer C: 5 centimeters
The hole has 25% of the original area; its radius is therefore √0.25 = 0.5 times 10, or 5 centimeters.

232. HSPT-M15-058 - Answer A: 58 square units
The original area is 54; removing the top-center cube removes one exposed face but reveals five neighboring faces, giving 54 − 1 + 5 = 58.

233. HSPT-M15-059 - Answer C: 77 minutes
The elapsed interval is 87 minutes; subtracting the included 10-minute intermission leaves 77 minutes of film.

234. HSPT-M15-060 - Answer C: 3 minutes 30 seconds
Delivering 36 liters requires 3 minutes of pumping, split into two 90-second runs with one 30-second pause between them.

235. HSPT-M15-061 - Answer B: 900 grams
Convert the width to 1.5 meters; the area is 1.5 × 2.4 = 3.6 square meters, and 3.6 × 250 = 900 grams.

236. HSPT-M15-062 - Answer B: 10
The day frequencies add to 6 + 3 + 1 = 10; days with zero delays still count as recorded days.

237. HSPT-M15-063 - Answer A: 8
The six equally likely awards average (0 + 0 + 0 + 2 + 2 + k)/6; setting this to 2 gives k = 8.

238. HSPT-M15-064 - Answer A: 10 through 19
In increasing order, positions 3 through 7 lie in the middle interval, so both middle readings, fifth and sixth, and their average lie there.

Language

239. HSPT-L15-001 - Answer A: Choose item (a) from the list.
The opening and closing parentheses must immediately surround the label a; enclosing additional words or omitting a mark fails that requirement.

240. HSPT-L15-002 - Answer B: "The lamps go out / the harbor waits"
A spaced slash marks a poetry line break when lines are quoted within prose; an ellipsis would instead suggest omitted material.

241. HSPT-L15-003 - Answer C: pages 18–24
The range dash connects the endpoints of an inclusive page span; a comma names separate pages rather than every intervening page.

242. HSPT-L15-004 - Answer B: I found the listing on eBay.
The supplied proper name uses a lowercase e followed by an uppercase B; its internal capitalization remains eBay within the sentence.

243. HSPT-L15-005 - Answer D: Night Routes: A Guide to the Back Roads
The subtitle begins with capital A even though it is an article; interior to and the remain lowercase while principal words are capitalized.

244. HSPT-L15-006 - Answer B: biscuit
Biscuit retains the sc and uit sequences; the alternatives substitute q, omit an i, or add an unnecessary ending.

245. HSPT-L15-007 - Answer C: idiosyncrasy
Idiosyncrasy begins with idio, uses y in syn, and ends in crasy rather than cracy; each alternative changes one of those elements.

246. HSPT-L15-008 - Answer C: phenomena; phenomenon
Phenomena is plural and follows several; phenomenon is singular and follows one. The two forms cannot be interchanged or used indiscriminately.

247. HSPT-L15-009 - Answer D: Should
Should the projector fail means if the projector should fail; should permits this conditional inversion with the base verb fail.

248. HSPT-L15-010 - Answer D: knows
Know describes a state here and normally takes the simple present. Knows also agrees with Zora; the other forms misuse aspect, agreement, or participles.

249. HSPT-L15-011 - Answer A: mitigate; militate
Mitigate means lessen a burden; militate against means operate as a factor opposing something, here participation despite the grant.

250. HSPT-L15-012 - Answer A: prescribe; proscribe
Prescribe means specify or require a practice; proscribe means forbid it. The opening is required, while promotional messages are banned.

251. HSPT-L15-013 - Answer D: Why Selected Older Words Still Belong in a Dictionary
The proposal retains selected older entries to aid understanding; it does not urge universal revival, replacement of modern words, or unlimited space.

252. HSPT-L15-014 - Answer A: A learner's dictionary should give current words most of its space, even if it retains a few older ones.
The sentence accepts priority for current vocabulary while allowing a limited older selection; the alternatives abandon the proposal or contradict its selection principle.

253. HSPT-L15-015 - Answer C: A limited selection of older entries can be useful when their value to readers justifies the space.
The revision makes inclusion selective and useful to readers, consistent with sentences 1 and 7, instead of granting every obsolete word equal space.

254. HSPT-L15-016 - Answer D: The guide asked, "Which path reaches the saddle first?"
The quoted words themselves form a question, so their question mark belongs inside the closing quotation mark.

255. HSPT-L15-017 - Answer B: "The ink," the engraver said, "will dry unevenly."
Commas mark the interruption, and will stays lowercase because the quotation continues the same sentence after the speaker identification.

256. HSPT-L15-018 - Answer B: From Earth, the probe carried a small sealed sample of earth.
Earth is capitalized as the planet's proper name; earth meaning soil is a common noun and remains lowercase.

257. HSPT-L15-019 - Answer A: They traveled south to study traditional crafts in the South.
South is capitalized when naming a recognized US region; south indicating a direction of travel remains lowercase.

258. HSPT-L15-020 - Answer D: cactus
Cactus is the standard spelling of the singular noun; the alternatives change its consonants or the final syllable.

259. HSPT-L15-021 - Answer B: sheathe
The verb meaning to put a blade into its protective covering is spelled sheathe, with ea and a final e.

260. HSPT-L15-022 - Answer D: exhilarate
Exhilarate, meaning to make someone feel excited or cheerful, uses ilar after exh and contains only one l.

261. HSPT-L15-023 - Answer B: drift
After let followed by an object, the next verb takes its base form without to: let the glider drift.

262. HSPT-L15-024 - Answer C: the platform remained level above the water
The opening phrase describes what is suspended, so the platform must be the subject immediately following that phrase.

263. HSPT-L15-025 - Answer A: Nobody knows why the clock stopped.
An embedded question uses statement word order and the ordinary past-tense verb stopped, rather than direct-question inversion with did.

264. HSPT-L15-026 - Answer C: With no stake in either side's success, the disinterested referee considered both claims carefully.
Having no stake in either side's success establishes impartiality; attentive consideration also distinguishes that meaning from boredom or lack of interest.

265. HSPT-L15-027 - Answer C: censor; censure
Censor means suppress restricted material; censure means formally express strong disapproval of someone's conduct.

266. HSPT-L15-028 - Answer D: Sentence 7
Sentence 7 concerns poster colors rather than testing sound equipment under realistic conditions, so it interrupts the passage's focus.

267. HSPT-L15-029 - Answer A: The operator listens for unwanted noise and checks whether ordinary speech is clear.
The revision retains both parallel checking actions without adding an unsupported cause, replacing one check, or making either check unnecessary.

268. HSPT-L15-030 - Answer A: After sentence 8
Sentence 8 introduces repeating the test without changing position; the added sentence explains that choice before sentence 9 warns against multiple changes.

269. HSPT-L15-031 - Answer A: the pilots' flight bags
The owners are plural pilots, so the possessive apostrophe follows their final s; bags is an ordinary plural without an apostrophe.

270. HSPT-L15-032 - Answer D: The channel was narrow; however, the barge passed safely.
A semicolon separates the independent clauses, and a comma follows however when it introduces the second clause.

271. HSPT-L15-033 - Answer C: The route includes Dover, Delaware; Salem, Oregon; and Helena, Montana.
Commas connect each city with its state, while semicolons separate the three list items that already contain commas.

272. HSPT-L15-034 - Answer D: The crew spoke Italian while crossing the canal.
Names of languages are proper nouns and are capitalized; crew, spoke, and canal are ordinary words in this sentence.

273. HSPT-L15-035 - Answer B: Renaissance art from the fifteenth century
Renaissance names a historical period and is capitalized; a general century description such as fifteenth century remains lowercase.

274. HSPT-L15-036 - Answer B: accordion
The standard US English spelling is accordion, with two c's and the ending ion rather than ian or eon.

275. HSPT-L15-037 - Answer A: millennium
Millennium, a period of one thousand years, contains both a double l and a double n before its final ium.

276. HSPT-L15-038 - Answer C: more smoothly
Smoothly forms its comparative with more; the sentence compares two conditions, so neither a double comparative nor a superlative fits.

277. HSPT-L15-039 - Answer A: are
All refers to the remaining channels, a plural subject; but one excludes an item without making the subject singular.

278. HSPT-L15-040 - Answer C: whoever
Whoever is the subject of has made within its clause; the entire clause, not that pronoun alone, follows the preposition to.

279. HSPT-L15-041 - Answer C: ascent
Ascent means an upward climb; assent means agreement, accent concerns pronunciation or emphasis, and access means a means of entry.

280. HSPT-L15-042 - Answer A: precedent; precedence
A precedent is an earlier example that guides a later decision; precedence means priority over a competing concern.

281. HSPT-L15-043 - Answer D: tortuous; torturous
Tortuous describes twisting or winding; torturous describes something causing severe discomfort, so the path and noise require different adjectives.

282. HSPT-L15-044 - Answer B: However
However introduces a limit on the preceding benefit: optional notes help readers, but essential steps must remain in the main explanation.

283. HSPT-L15-045 - Answer B: A useful footnote adds depth while leaving a complete route through the main explanation.
The conclusion combines the note's optional added detail with the requirement that the main explanation remain understandable without it.

284. HSPT-L15-046 - Answer C: To create the tint, mix red, white, and a touch of blue.
A comma closes the introductory phrase, and commas separate the three objects being mixed without separating the verb from its objects.

285. HSPT-L15-047 - Answer C: The stencil requires three tools: a knife, a mat, and a ruler.
The correct colon follows a complete clause; the other versions put it directly between a verb and the objects or complement it needs.

286. HSPT-L15-048 - Answer D: The ferry runs on Labor Day but not every Monday.
Labor Day is a holiday name and Monday is a day name, so both are capitalized; ferry remains a common noun.

287. HSPT-L15-049 - Answer B: They crossed the Strait of Gibraltar before entering the Mediterranean Sea.
The full geographic names capitalize Strait, Gibraltar, Mediterranean, and Sea; the connecting preposition of remains lowercase.

288. HSPT-L15-050 - Answer D: pavilion
Pavilion is spelled with one l and the ending ion; the alternatives add a letter or omit required vowels.

289. HSPT-L15-051 - Answer B: pistachio
Pistachio, the name of the nut, includes ch and ends in io rather than oe or iao.

290. HSPT-L15-052 - Answer A: zucchini
Zucchini contains two c's followed by h and only one n; the alternatives omit or double the wrong consonant.

291. HSPT-L15-053 - Answer D: fallen
Had requires a past participle; fallen is the past participle of fall, whereas fell is its simple past form.

292. HSPT-L15-054 - Answer B: is
The fraction refers to cement, an uncountable substance, so it takes singular agreement: two thirds of the cement is wet.

293. HSPT-L15-055 - Answer C: Only after the bridge opened did traffic resume.
Fronted Only after requires auxiliary-subject inversion in the main clause: did comes before traffic, while resume follows the subject.

294. HSPT-L15-056 - Answer A: serial; cereal
A serial appears in installments; cereal is a grain-based food commonly eaten with milk, so the two contexts require different homophones.

295. HSPT-L15-057 - Answer D: dairy; diary
A dairy handles milk or milk products, while a diary is a personal record of daily events.

296. HSPT-L15-058 - Answer A: gamut; gambit
Gamut means a complete range; gambit means an opening move or calculated maneuver intended to gain an advantage.

297. HSPT-L15-059 - Answer C: Returning to a familiar image in a changed state can give a story's ending a sense of completion.
The passage explains how a changed closing image connects with an opening image; the pottery details illustrate that general writing technique.

298. HSPT-L15-060 - Answer A: 3, 4, 2
Sentence 3 establishes the initial clay, sentence 4 presents the later mark, and sentence 2 interprets that mark by comparison.